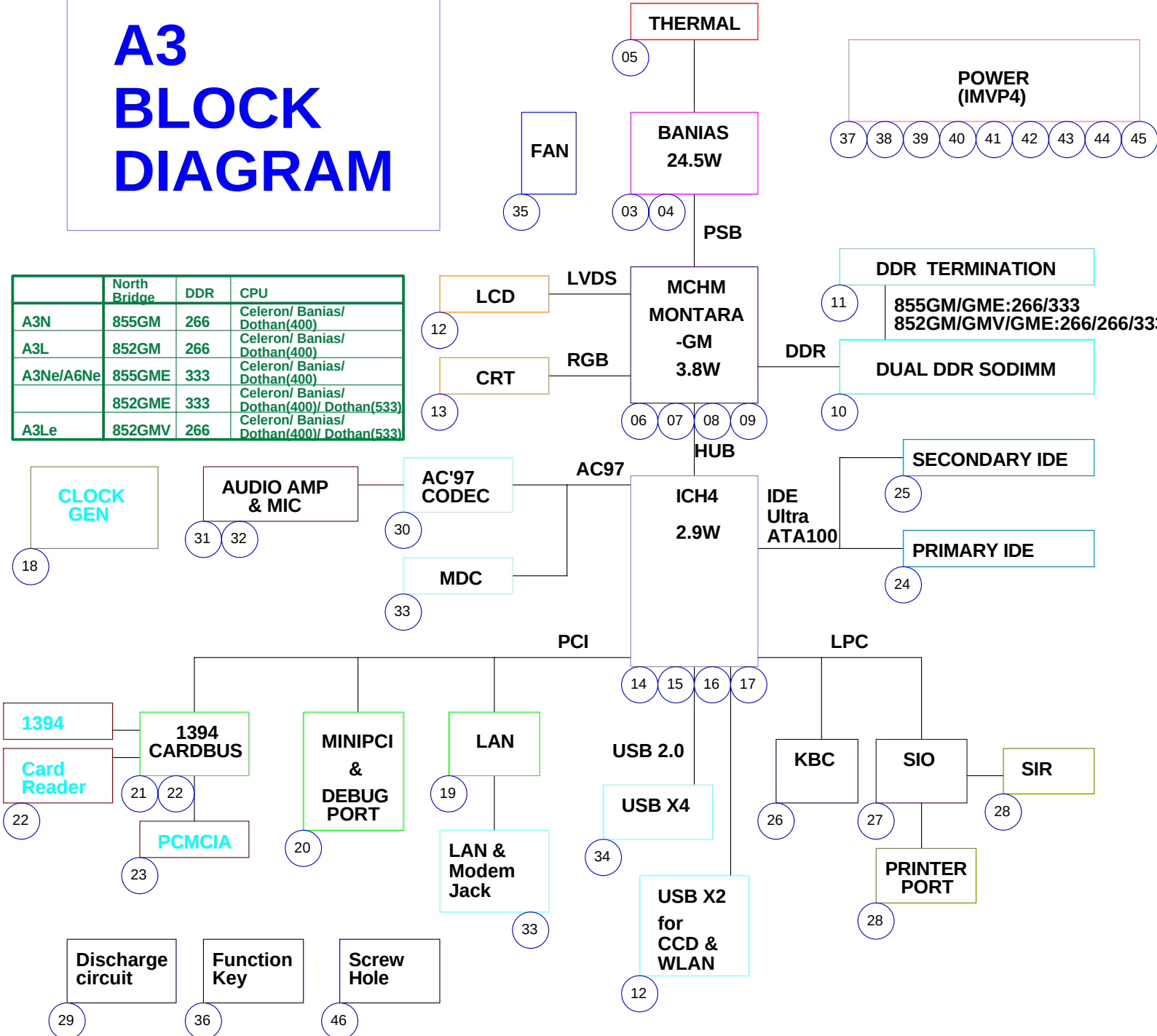


A3 BLOCK DIAGRAM

	North Bridge	DDR	CPU
A3N	855GM	266	Celeron/ Banias/ Dothan(400)
A3L	852GM	266	Celeron/ Banias/ Dothan(400)
A3Ne/A6Ne	855GME	333	Celeron/ Banias/ Dothan(400)
	852GME	333	Celeron/ Banias/ Dothan(400)/ Dothan(533)
A3Le	852GMV	266	Celeron/ Banias/ Dothan(400)/ Dothan(533)



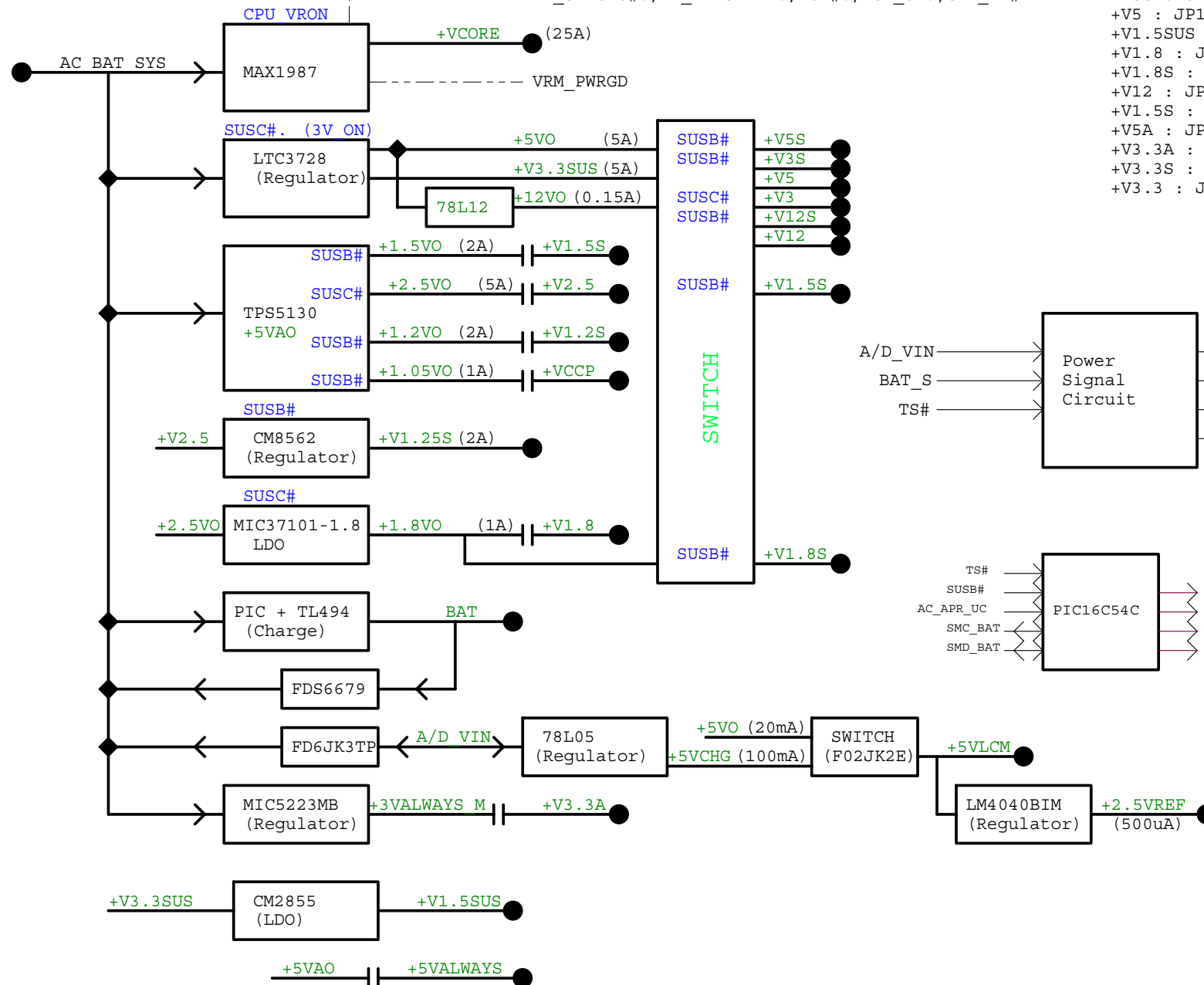
FILE LIST

01_BLOCK DIAGRAM
02_POWER DIAGRAM
03_CPU-BANIAS(HOST)
04_CPU-BANIAS(PWR)
05_THERMAL
06_NB-MCHM(DDR)
07_NB-MCHM(HOST)
08_NB-MCHM(VGA)
09_NB-MCHM(PWR)
10_DUAL DDR SODIMM
11_DDR TERMINATION
12_LVDS & BACKLIGHT
13_CRT CONNECTOR
14_ICH4-M(HUB_PCI)
15_ICH4-M(H_U_IDE_PM)
16_ICH4-M(PWR)
17_ICH4-M_PULLUP
18_CLOCK-ICS950815
19_LAN-RTL8100CL
20_MINIPCI
21_CB1394-R5C593(1)
22_CB1394-R5C593(2)
23_PCMCIA SOCKET
24_IDE-HD
25_IDE-ODD
26_KBC-M38857
27_SIO-ITE7805
28_IR&LPT_PORT
29_DISCHARGE CIRCUIT
30_CODEC-ALC650
31_AUDIO AMP
32_MIC
33_MDC&RJ45&RJ11
34_USB
35_FAN&Audio DJ
36_FUNCTION KEY
37_PWR & RESET SEQ
38_VCORE
39_1.25V&1.8V
40_2.5V&1.5V&1.35V&1.05V
41_SYSTEM
42_LOAD SWITCH
43_CHARGER
44_PIC16C54
45_BATLOW/SD#
46_SCREW HOLE & M/B SETTING
47_REVISION(1)
48_REVISION(2)
49_REVISION(3)
50_REVISION(4)
51_BLOCK DIAGRAM 5E
52_BLOCK DIAGRAM 7H
53_Part Reference 1
54_Part Reference 2
55_Power Net Reference

Battery in : 16.8 ~ 11.6V

```
PM STPCPU#., PM DPRSLPVR., PCI#., MCH OK., CLK EN#
```

The block diagram shows the PIC16C54C microcontroller. On the left, there are five input pins: TS#, SUSB#, AC_APR_UC, SMC_BAT, and SMD_BAT. On the right, there are four output pins: CHG_EN#, CHG_LED_UP, PWR_LED_UP, and BAT_LLOW.



POWER DIAGRAM

U3B

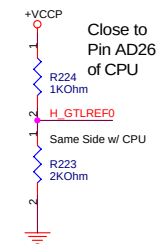
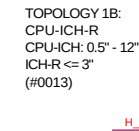
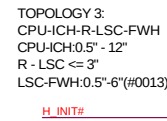
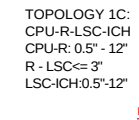
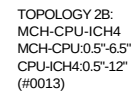
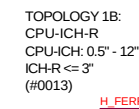
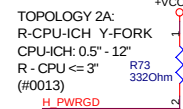
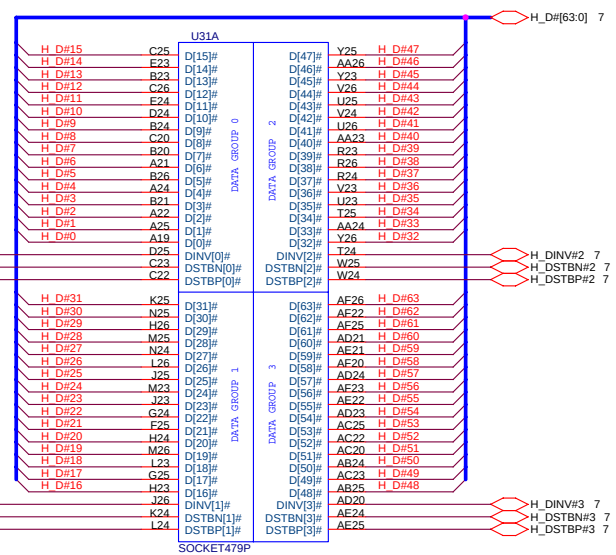
ADDRESS GROUP 0

ADDRESS GROUP 1

SOCKET479P

Pinout Details:

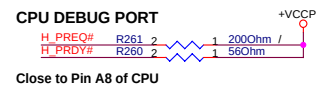
- Pin 1:** H_A# [16:3] (Red)
- Pin 2:** H_A#16 (Red)
- Pin 3:** H_A#15 (Red)
- Pin 4:** H_A#14 (Red)
- Pin 5:** H_A#13 (Red)
- Pin 6:** H_A#12 (Red)
- Pin 7:** H_A#11 (Red)
- Pin 8:** H_A#10 (Red)
- Pin 9:** H_A#9 (Red)
- Pin 10:** H_A#8 (Red)
- Pin 11:** H_A#7 (Red)
- Pin 12:** H_A#6 (Red)
- Pin 13:** H_A#5 (Red)
- Pin 14:** H_A#4 (Red)
- Pin 15:** H_A#3 (Red)
- Pin 16:** H_REQ#4 (Red)
- Pin 17:** H_REQ#3 (Red)
- Pin 18:** H_REQ#2 (Red)
- Pin 19:** H_REQ#1 (Red)
- Pin 20:** H_REQ#0 (Red)
- Pin 21:** H_A# [31:17] (Red)
- Pin 22:** H_A#31 (Red)
- Pin 23:** H_A#30 (Red)
- Pin 24:** H_A#29 (Red)
- Pin 25:** H_A#28 (Red)
- Pin 26:** H_A#27 (Red)
- Pin 27:** H_A#26 (Red)
- Pin 28:** H_A#25 (Red)
- Pin 29:** H_A#24 (Red)
- Pin 30:** H_A#23 (Red)
- Pin 31:** H_A#22 (Red)
- Pin 32:** H_A#21 (Red)
- Pin 33:** H_A#20 (Red)
- Pin 34:** H_A#19 (Red)
- Pin 35:** H_A#18 (Red)
- Pin 36:** H_A#17 (Red)
- Pin 37:** H_A#16 (Red)
- Pin 38:** H_A#15 (Red)
- Pin 39:** H_A#14 (Red)
- Pin 40:** H_A#13 (Red)
- Pin 41:** H_A#12 (Red)
- Pin 42:** H_A#11 (Red)
- Pin 43:** H_A#10 (Red)
- Pin 44:** H_A#9 (Red)
- Pin 45:** H_A#8 (Red)
- Pin 46:** H_A#7 (Red)
- Pin 47:** H_A#6 (Red)
- Pin 48:** H_A#5 (Red)
- Pin 49:** H_A#4 (Red)
- Pin 50:** H_A#3 (Red)
- Pin 51:** H_A#2 (Red)
- Pin 52:** H_A#1 (Red)
- Pin 53:** H_A#0 (Red)
- Pin 54:** H_A#-1 (Red)
- Pin 55:** H_A#-2 (Red)
- Pin 56:** H_A#-3 (Red)
- Pin 57:** H_A#-4 (Red)
- Pin 58:** H_A#-5 (Red)
- Pin 59:** H_A#-6 (Red)
- Pin 60:** H_A#-7 (Red)
- Pin 61:** H_A#-8 (Red)
- Pin 62:** H_A#-9 (Red)
- Pin 63:** H_A#-10 (Red)
- Pin 64:** H_A#-11 (Red)
- Pin 65:** H_A#-12 (Red)
- Pin 66:** H_A#-13 (Red)
- Pin 67:** H_A#-14 (Red)
- Pin 68:** H_A#-15 (Red)
- Pin 69:** H_A#-16 (Red)
- Pin 70:** H_A#-17 (Red)
- Pin 71:** H_A#-18 (Red)
- Pin 72:** H_A#-19 (Red)
- Pin 73:** H_A#-20 (Red)
- Pin 74:** H_A#-21 (Red)
- Pin 75:** H_A#-22 (Red)
- Pin 76:** H_A#-23 (Red)
- Pin 77:** H_A#-24 (Red)
- Pin 78:** H_A#-25 (Red)
- Pin 79:** H_A#-26 (Red)
- Pin 80:** H_A#-27 (Red)
- Pin 81:** H_A#-28 (Red)
- Pin 82:** H_A#-29 (Red)
- Pin 83:** H_A#-30 (Red)
- Pin 84:** H_A#-31 (Red)
- Pin 85:** H_A#-32 (Red)
- Pin 86:** H_A#-33 (Red)
- Pin 87:** H_A#-34 (Red)
- Pin 88:** H_A#-35 (Red)
- Pin 89:** H_A#-36 (Red)
- Pin 90:** H_A#-37 (Red)
- Pin 91:** H_A#-38 (Red)
- Pin 92:** H_A#-39 (Red)
- Pin 93:** H_A#-40 (Red)
- Pin 94:** H_A#-41 (Red)
- Pin 95:** H_A#-42 (Red)
- Pin 96:** H_A#-43 (Red)
- Pin 97:** H_A#-44 (Red)
- Pin 98:** H_A#-45 (Red)
- Pin 99:** H_A#-46 (Red)
- Pin 100:** H_A#-47 (Red)
- Pin 101:** H_A#-48 (Red)
- Pin 102:** H_A#-49 (Red)
- Pin 103:** H_A#-50 (Red)
- Pin 104:** H_A#-51 (Red)
- Pin 105:** H_A#-52 (Red)
- Pin 106:** H_A#-53 (Red)
- Pin 107:** H_A#-54 (Red)
- Pin 108:** H_A#-55 (Red)
- Pin 109:** H_A#-56 (Red)
- Pin 110:** H_A#-57 (Red)
- Pin 111:** H_A#-58 (Red)
- Pin 112:** H_A#-59 (Red)
- Pin 113:** H_A#-60 (Red)
- Pin 114:** H_A#-61 (Red)
- Pin 115:** H_A#-62 (Red)
- Pin 116:** H_A#-63 (Red)
- Pin 117:** H_A#-64 (Red)
- Pin 118:** H_A#-65 (Red)
- Pin 119:** H_A#-66 (Red)
- Pin 120:** H_A#-67 (Red)
- Pin 121:** H_A#-68 (Red)
- Pin 122:** H_A#-69 (Red)
- Pin 123:** H_A#-70 (Red)
- Pin 124:** H_A#-71 (Red)
- Pin 125:** H_A#-72 (Red)
- Pin 126:** H_A#-73 (Red)
- Pin 127:** H_A#-74 (Red)
- Pin 128:** H_A#-75 (Red)
- Pin 129:** H_A#-76 (Red)
- Pin 130:** H_A#-77 (Red)
- Pin 131:** H_A#-78 (Red)
- Pin 132:** H_A#-79 (Red)
- Pin 133:** H_A#-80 (Red)
- Pin 134:** H_A#-81 (Red)
- Pin 135:** H_A#-82 (Red)
- Pin 136:** H_A#-83 (Red)
- Pin 137:** H_A#-84 (Red)
- Pin 138:** H_A#-85 (Red)
- Pin 139:** H_A#-86 (Red)
- Pin 140:** H_A#-87 (Red)
- Pin 141:** H_A#-88 (Red)
- Pin 142:** H_A#-89 (Red)
- Pin 143:** H_A#-90 (Red)
- Pin 144:** H_A#-91 (Red)
- Pin 145:** H_A#-92 (Red)
- Pin 146:** H_A#-93 (Red)
- Pin 147:** H_A#-94 (Red)
- Pin 148:** H_A#-95 (Red)
- Pin 149:** H_A#-96 (Red)
- Pin 150:** H_A#-97 (Red)
- Pin 151:** H_A#-98 (Red)
- Pin 152:** H_A#-99 (Red)
- Pin 153:** H_A#-100 (Red)
- Pin 154:** H_A#-101 (Red)
- Pin 155:** H_A#-102 (Red)
- Pin 156:** H_A#-103 (Red)
- Pin 157:** H_A#-104 (Red)
- Pin 158:** H_A#-105 (Red)
- Pin 159:** H_A#-106 (Red)
- Pin 160:** H_A#-107 (Red)
- Pin 161:** H_A#-108 (Red)
- Pin 162:** H_A#-109 (Red)
- Pin 163:** H_A#-110 (Red)
- Pin 164:** H_A#-111 (Red)
- Pin 165:** H_A#-112 (Red)
- Pin 166:** H_A#-113 (Red)
- Pin 167:** H_A#-114 (Red)
- Pin 168:** H_A#-115 (Red)
- Pin 169:** H_A#-116 (Red)
- Pin 170:** H_A#-117 (Red)
- Pin 171:** H_A#-118 (Red)
- Pin 172:** H_A#-119 (



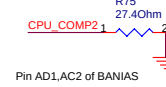
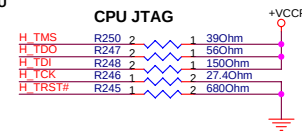
H_GTLREF0
LENGTH <=0.5"
WIDTH = 5 mils
SPACE >= 25 mils
X BPSB(#0004)

CPU_COMP2 :
Length <= 0.5"
Width = 20 mils(L1/L4)
Space >= 25 mils
X BPSB(#0001)

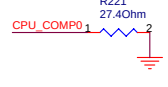
CPU_COMP0 :
Length <= 0.5"
Width = 20 mils(L1/L4)
Space >= 25 mils
X BPSB(#0001)



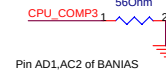
Close to Pin A12 of CPU
Width= 5 mils
Length <= 2"



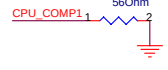
CPU_COMP3 :
Length <= 0.5"
Width = 5 mils
Space >= 25 mils
X BPSB(#0001)



CPU_COMP1 :
Length <= 0.5"
Width = 5 mils
Space >= 25 mils
X BPSB(#0001)

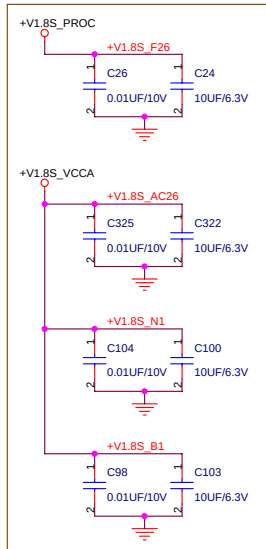


Pin AD1,AC2 of BANIAS

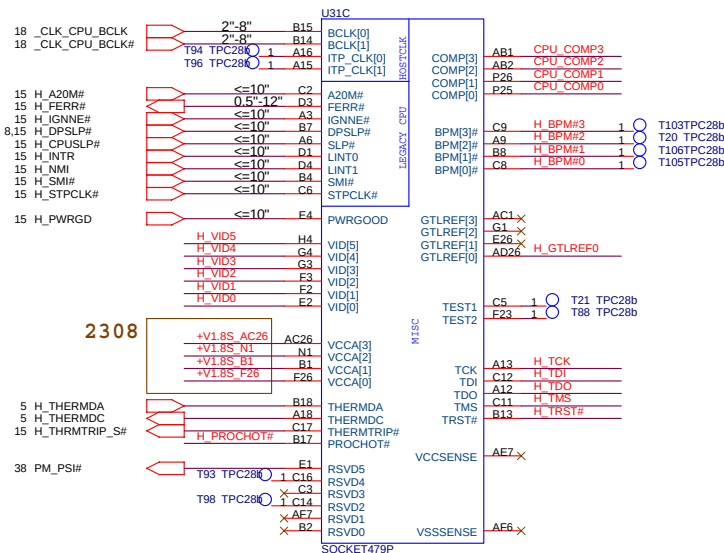


CPU-BANIAS(HOST)

1.71V - 1.89V(+/- 5%)
S0-S1M: 0.3A

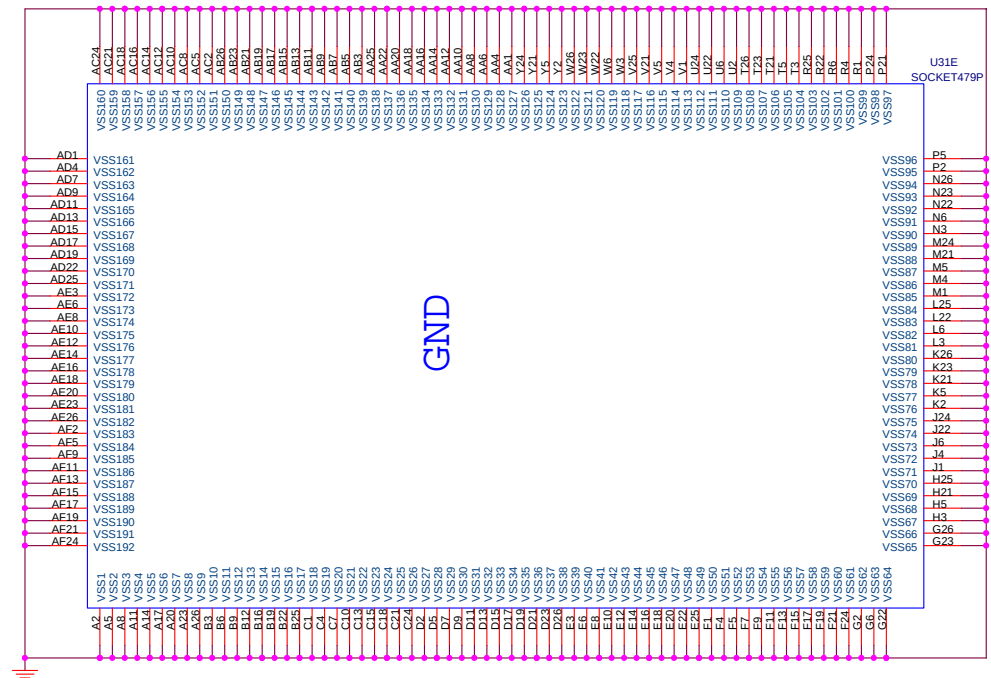
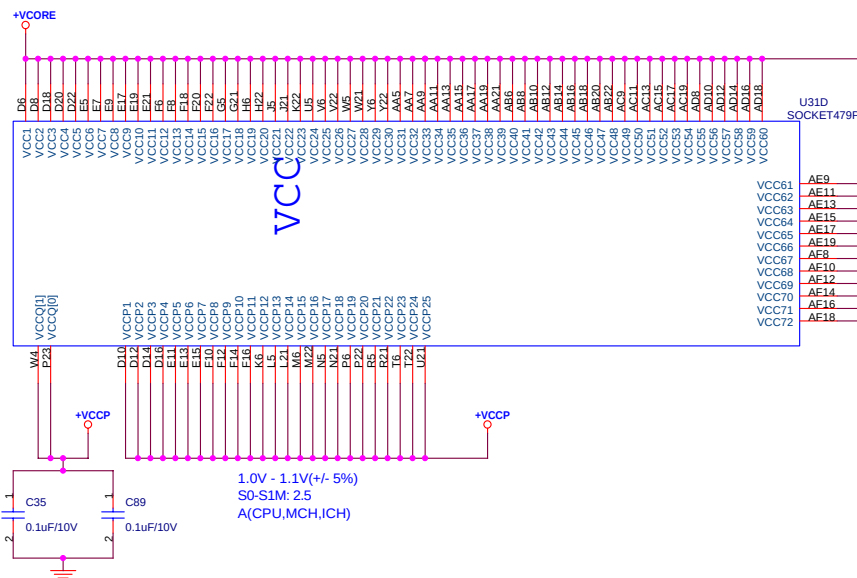


2025	2308
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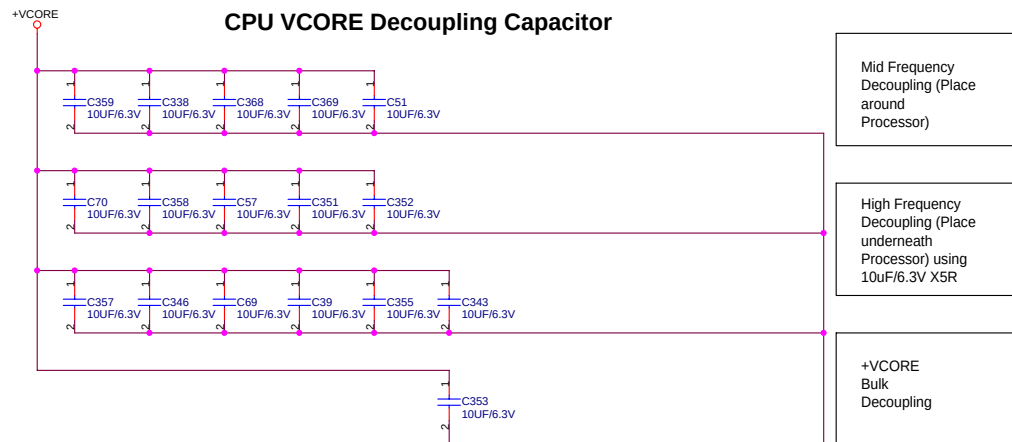


	Celeron	Banias	Dothan (400)	Dothan (533)
Frequency	100	100	100	133
VCCA[1:3]	1.8V	1.8V	1.8V	NC
VCCA[0]	1.8V	1.8V	1.8V	1.5V
U54 switch to	Pin 3,4			Pin 1,2

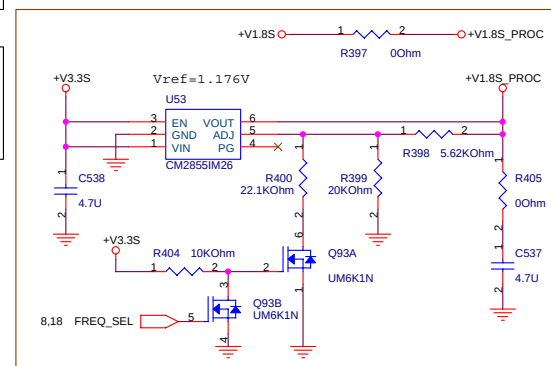
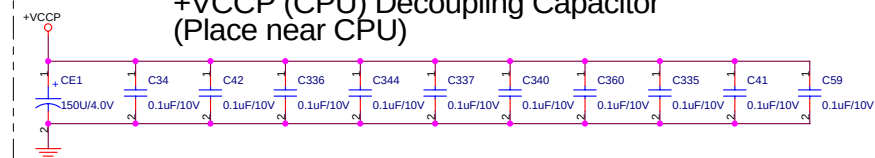
0.745V - 1.356V(+/- 1.5%)
C0: 25 A
C3: 7.59A
C4: 0.9A



CPU VCORE Decoupling Capacitor



+VCCP (CPU) Decoupling Capacitor
(Place near CPU)



2308

**For A3N/A3NE/A3L:
Load R397**

For A3LE:
Load C537, C538, Q93, R398,
R399, R400, R404, R405, U53

CPU-BANIAS(PWR)

Route H_THERMDA and H_THERMDC
on the same layer

-----OTHER SIGNALS

12 mils

=====GND

10 mils

=====H_THERMDA(10 mils)

10 mils

=====H_THERMDC(10 mils)

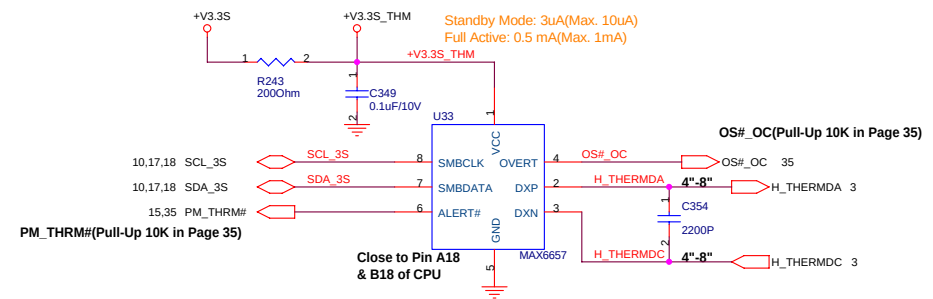
10 mils

=====GND

12 mils

-----OTHER SIGNALS

Avoid BPSB,Power



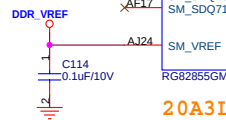
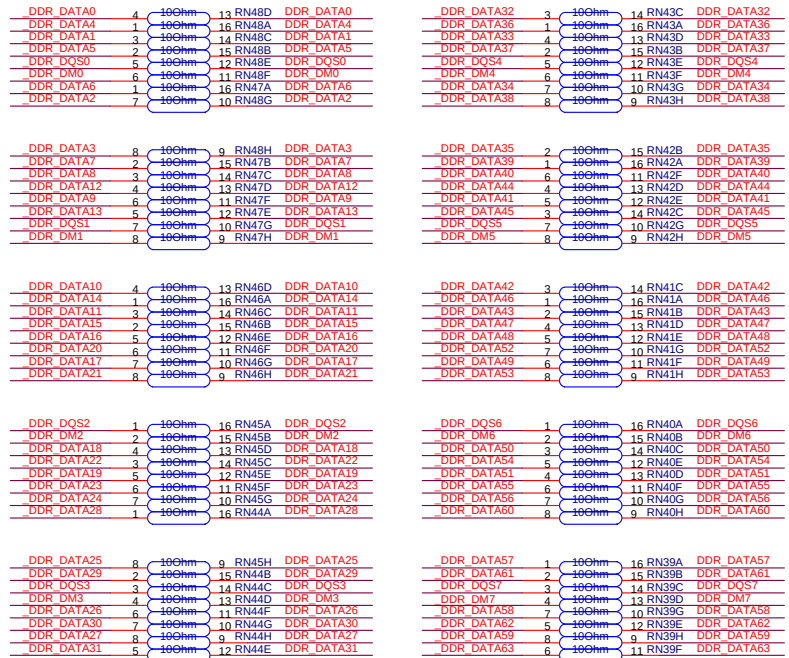
THERMAL

3.8W

Thermal Power: ~3.8W

LxWxH=37.5x37.5x2.58

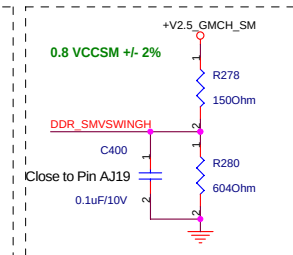
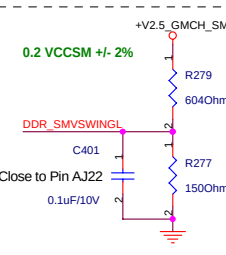
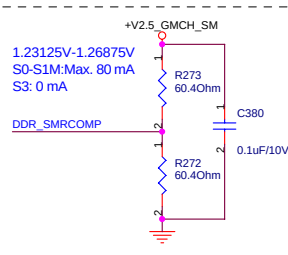
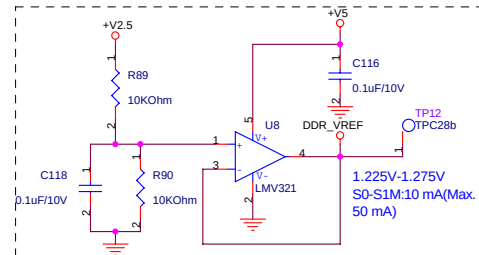
1107



20A3L

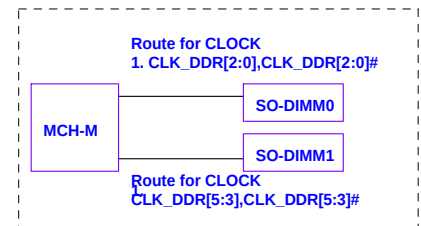
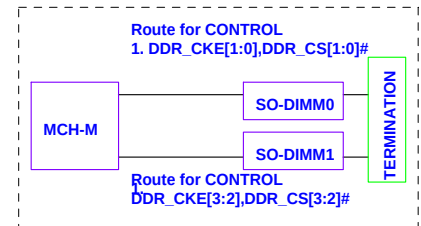
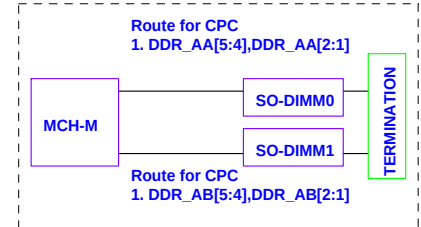
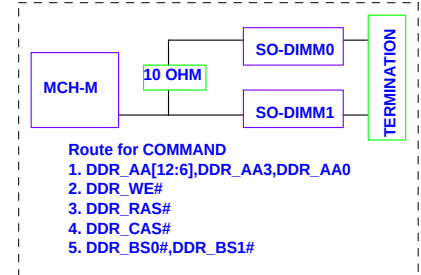
1.23125V-1.26875V
SO-S1M:Max. 80 mA
S3: 0 mA

Intel suggested that DDR_VREF should be turned off in S3-S5. But measure the leakage because there is no +V2.5S.

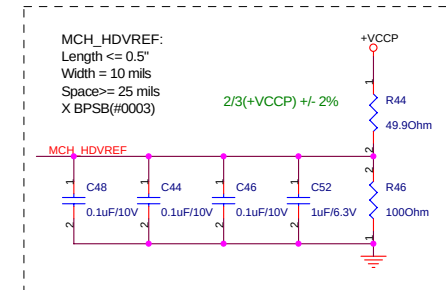
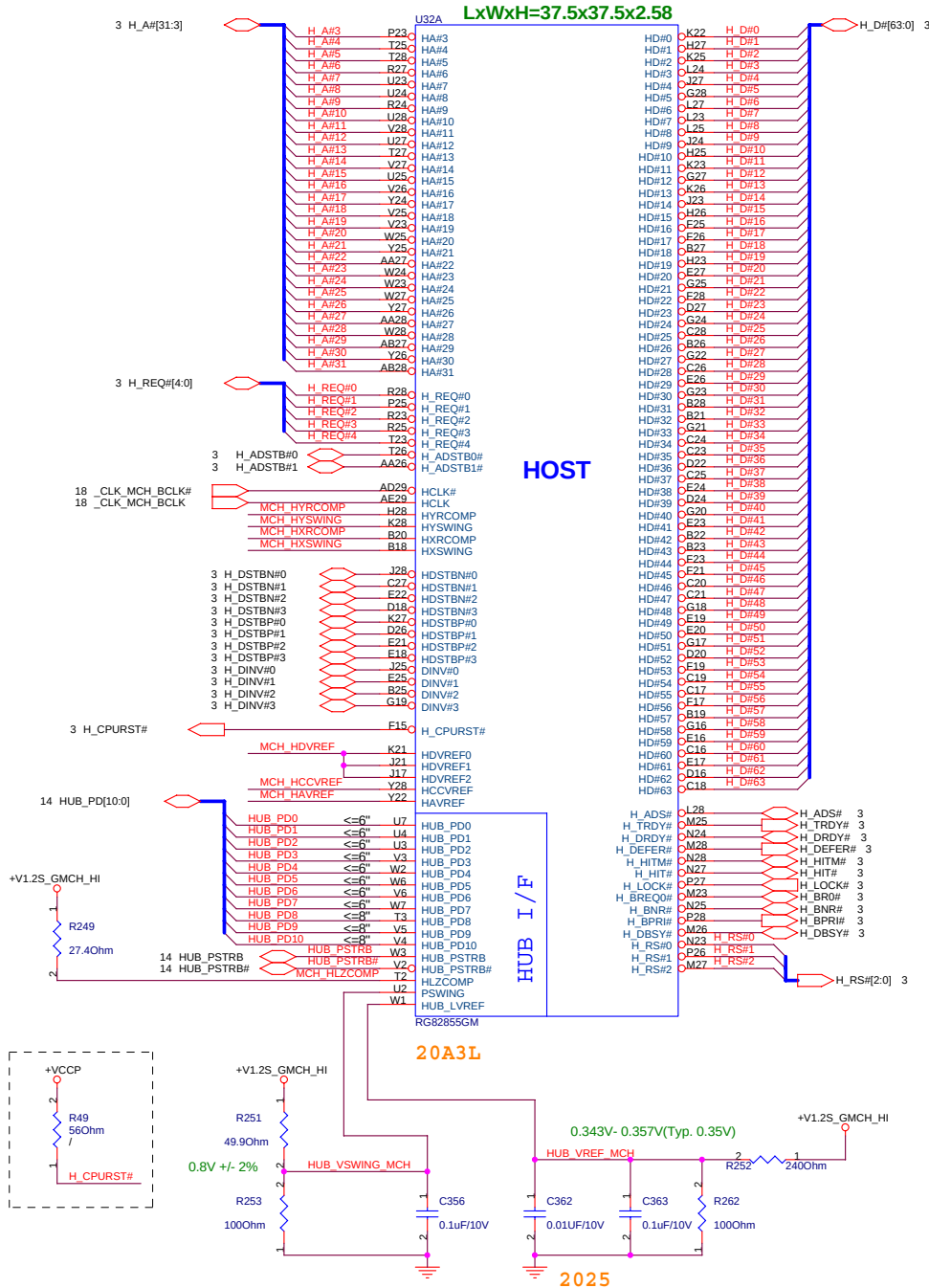
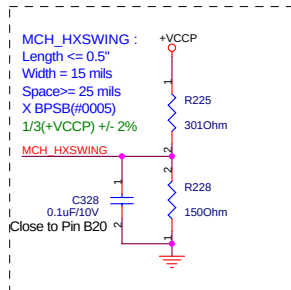


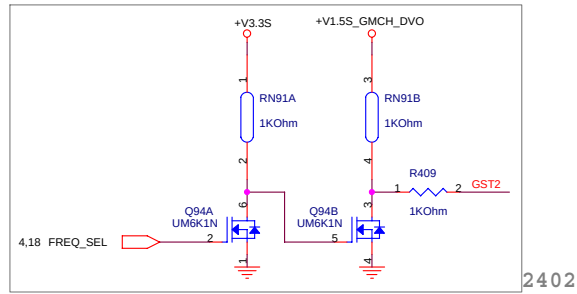
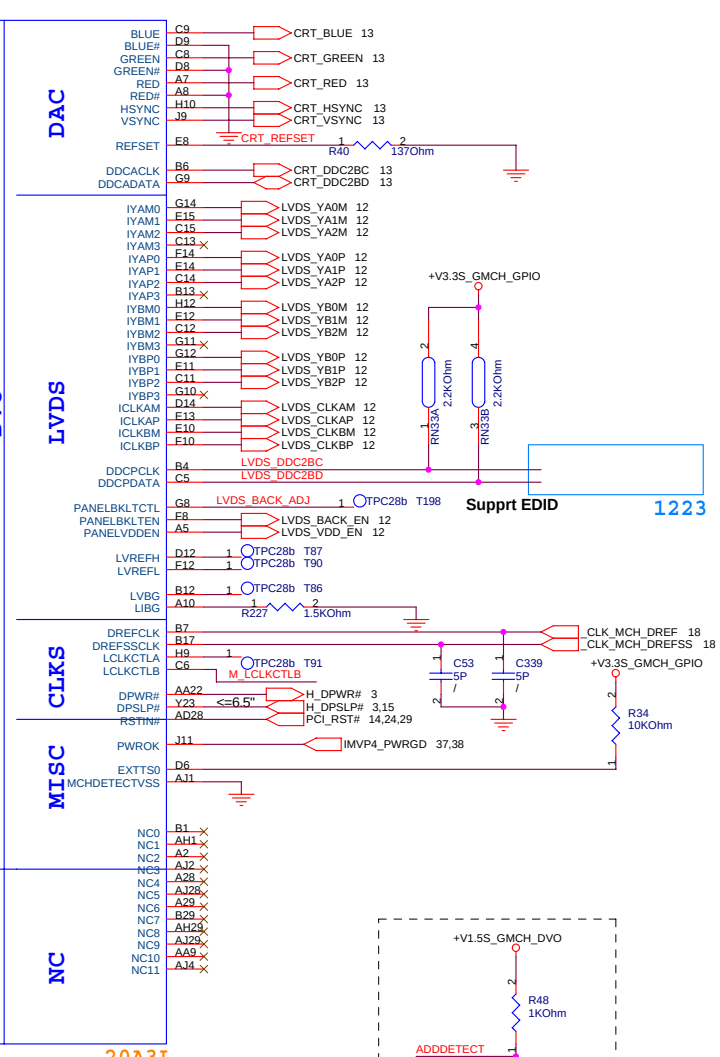
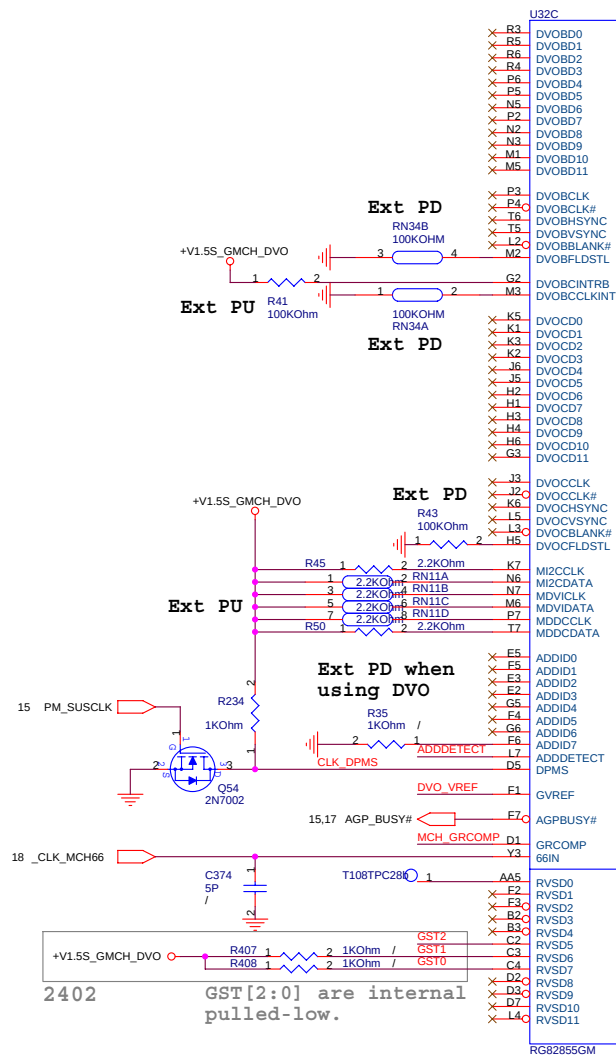
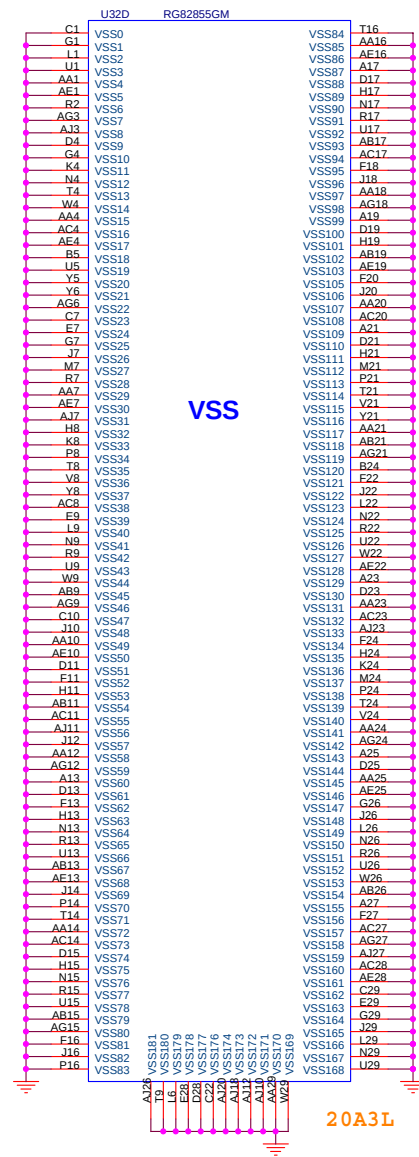
DDR_DATA[63:0] 10.11
DDR_DM[7:0] 10.11
DDR_DQS[7:0] 10.11

(MCH-Sighting041)
M-GM system memory interface generates single pulse CKE events which may cause intermittent hangs and display corruptions when using Micron and Infineon SO-DIMMs.

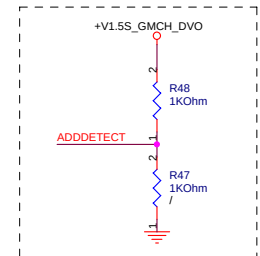
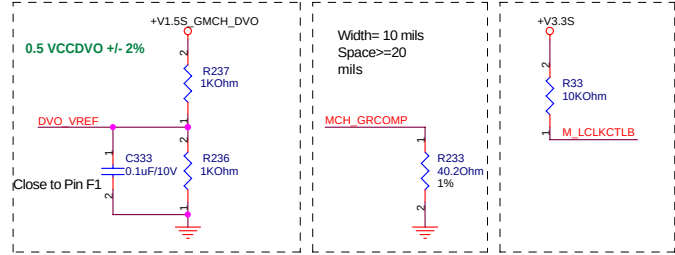


NB-MCHM(DDR)

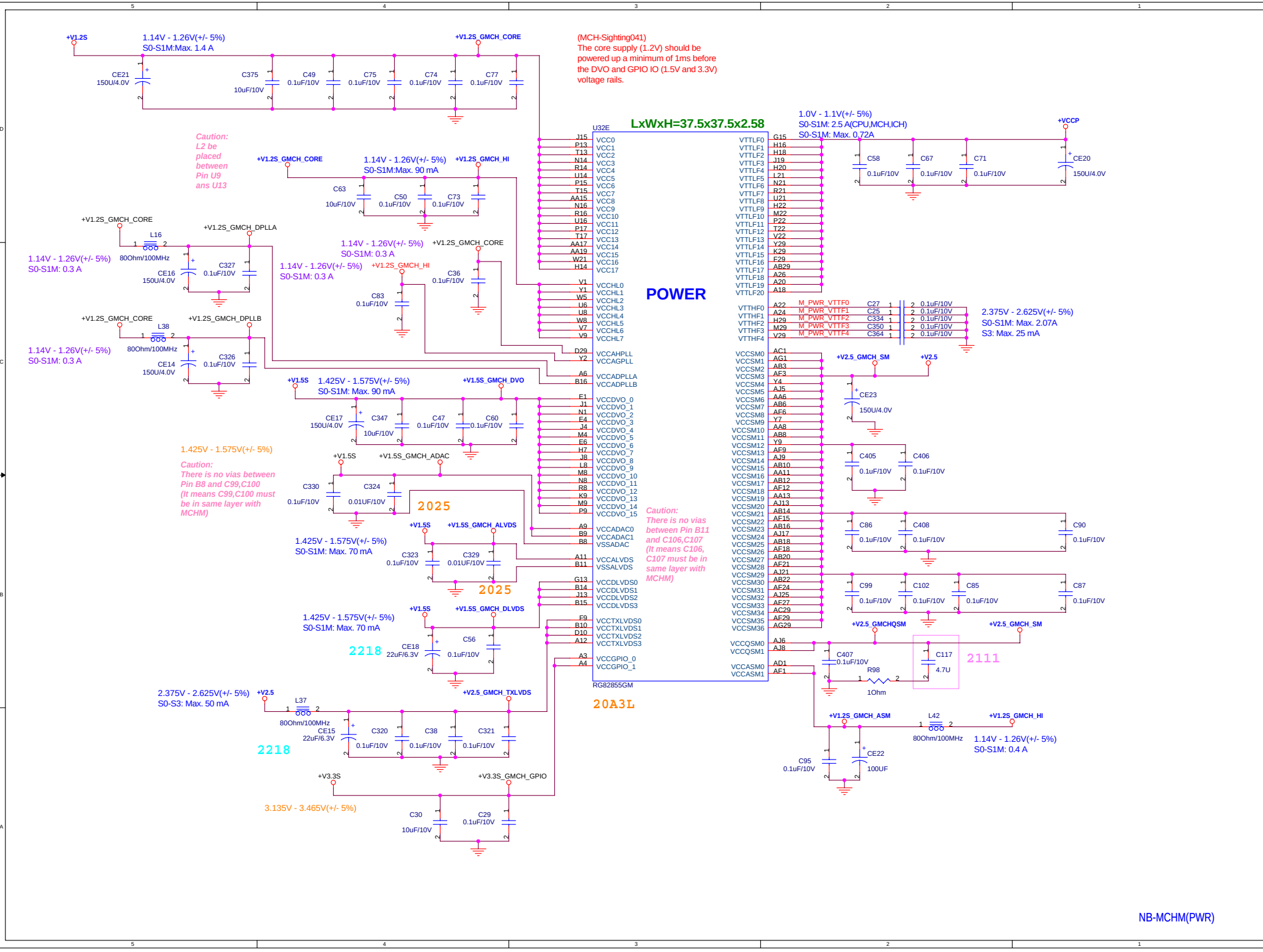




When no using DVO



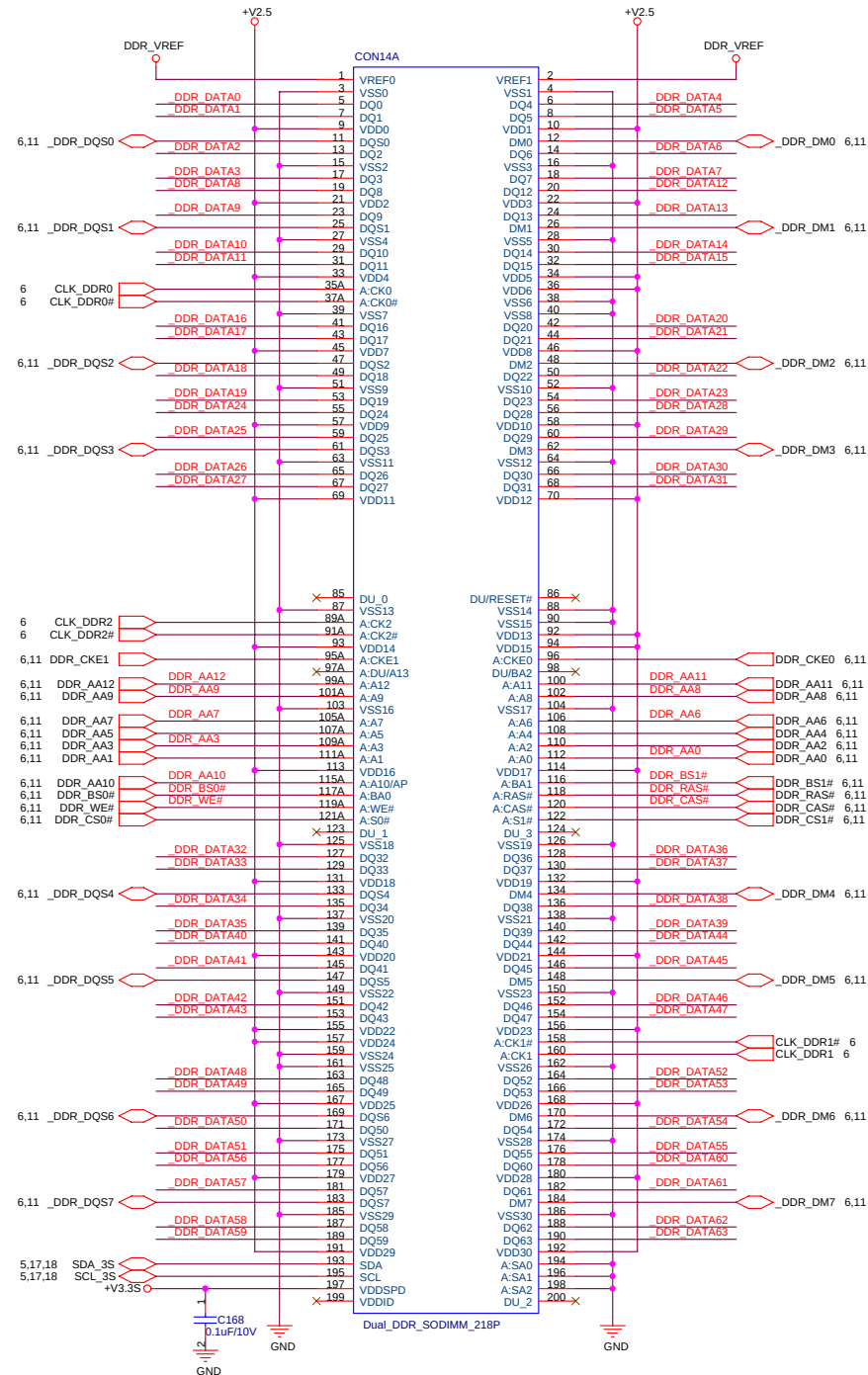
NB-MCHM(VGA)



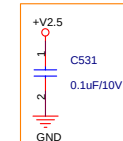
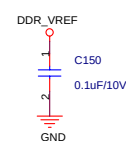
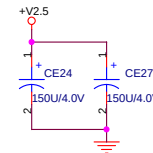
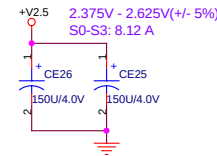
(MCH-Sighting041)
The core supply (1.2V) should be powered up a minimum of 1ms before the DVO and GPIO IO (1.5V and 3.3V) voltage rails.

POWER

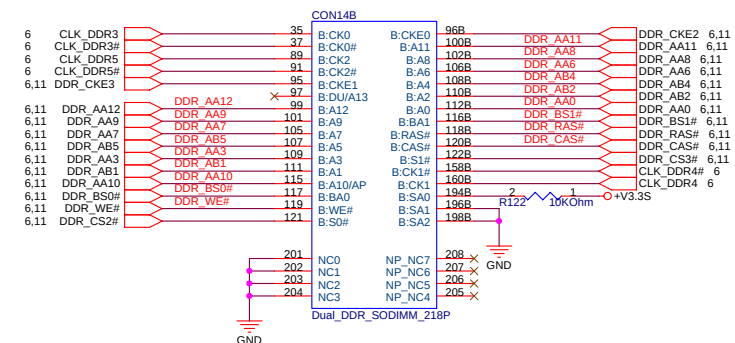
6.11 _DDR_DATA[63:0]



FOR +V2.5 DECOUPLING



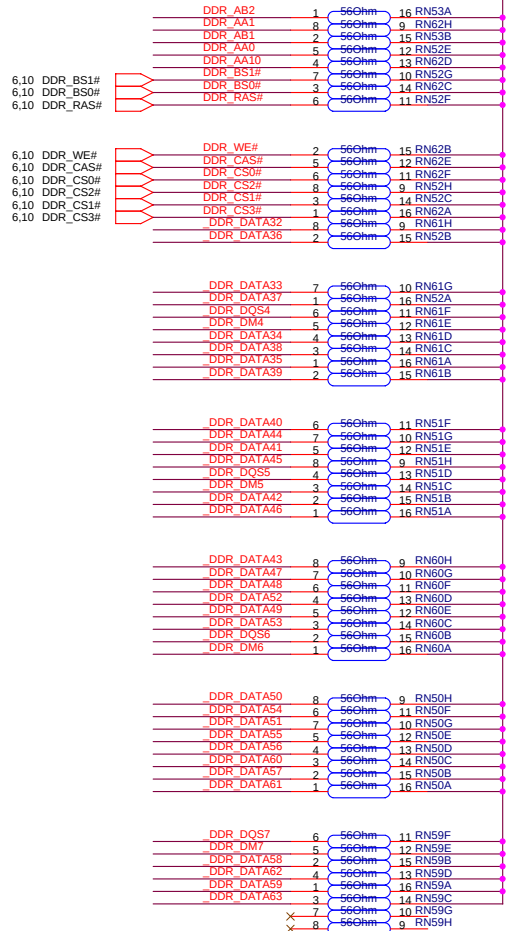
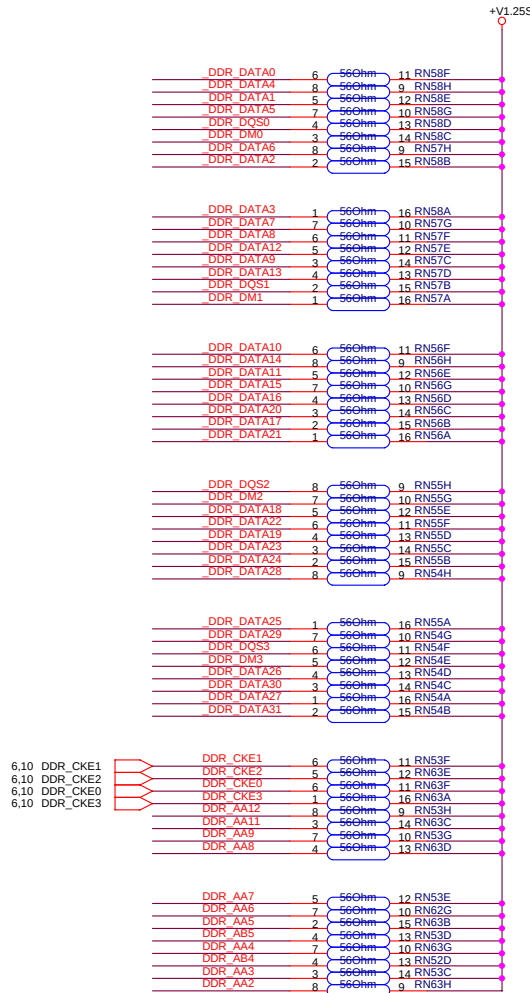
2024
EMI : Close to DDR
socket power plane
of +V2.5



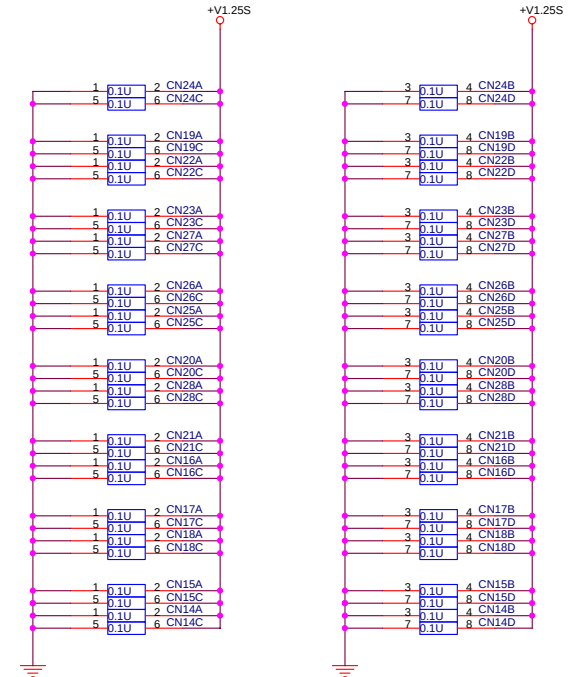
DUAL DDR SODIMM

DDR TERMINATION

6,10 _DDR_DATA[63:0]
6,10 _DDR_DM[8:0]
6,10 _DDR_DQS[8:0]
6,10 _DDR_AA[12:0]
6,10 _DDR_AB[2:1]
6,10 _DDR_AB[5:4]

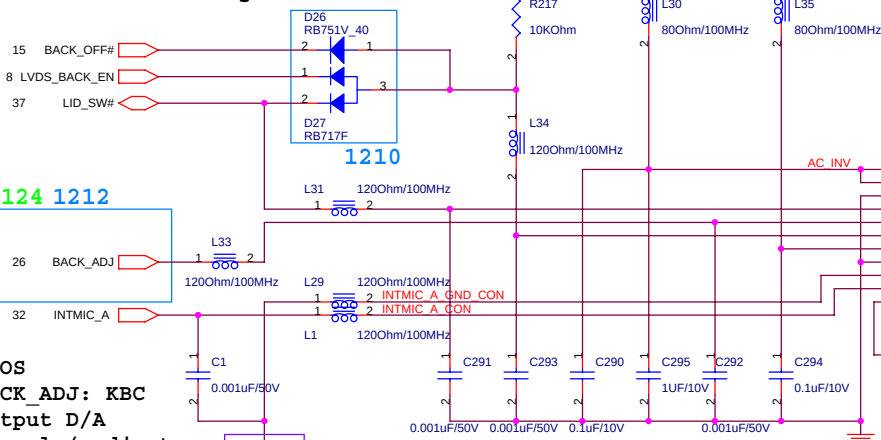


FOR +V1.25S DECOUPLING



DDR TERMINATION

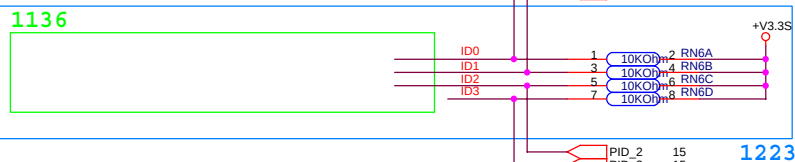
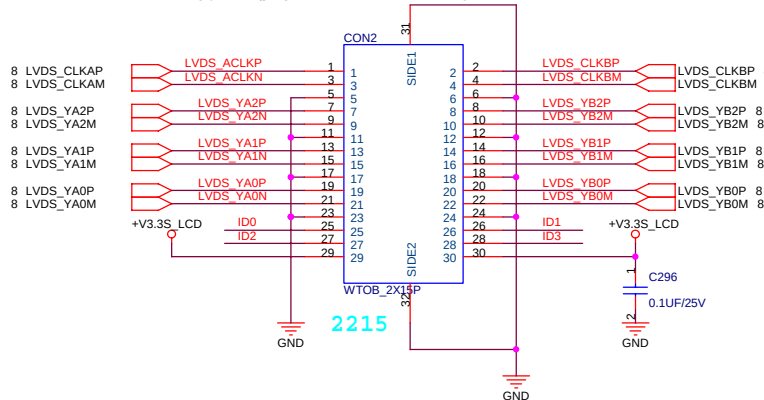
BIOS
BACK_OFF#:When user push "Fn+F7"
button, BIOS active this pin to
turn off back light.



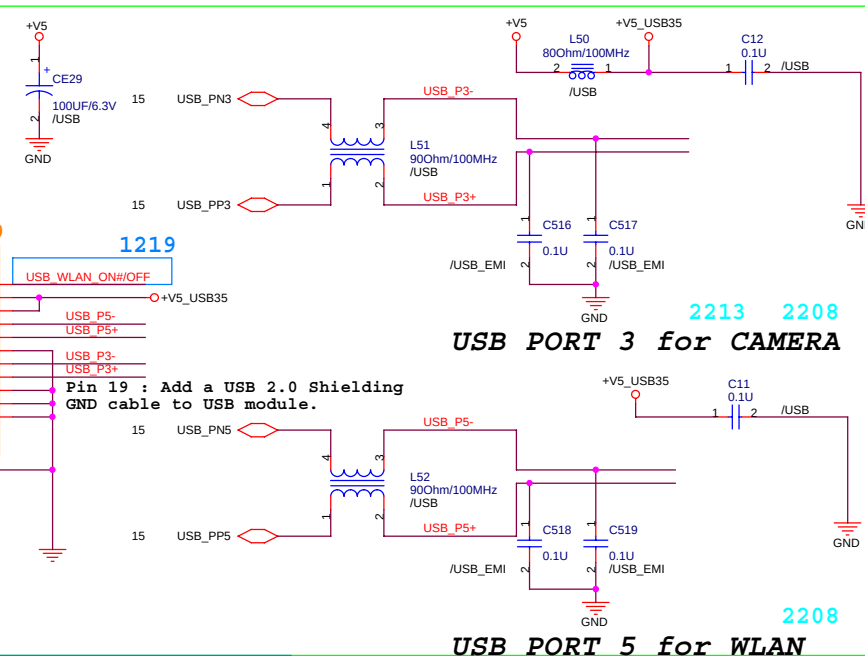
BIOS
BACK_ADJ: KBC
output D/A
signal (adjust
voltage level)
to adjust Back
light.

A3N use D1 R:1.0 Inverter Board

LCD CABLE ID:	PID3	PID2	PID1	PID0
14.1 XGA	1	1	0	1
15.1 XGA	1	1	0	1
15.1 SXGA+	1	0	1	1



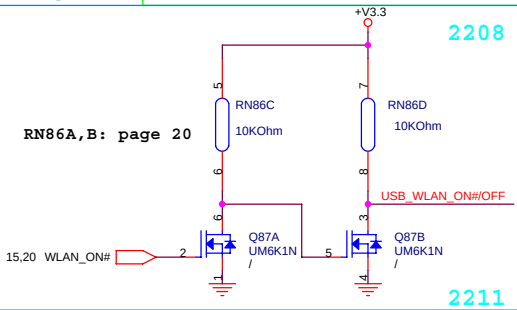
1128



USB PORT 3 for CAMERA

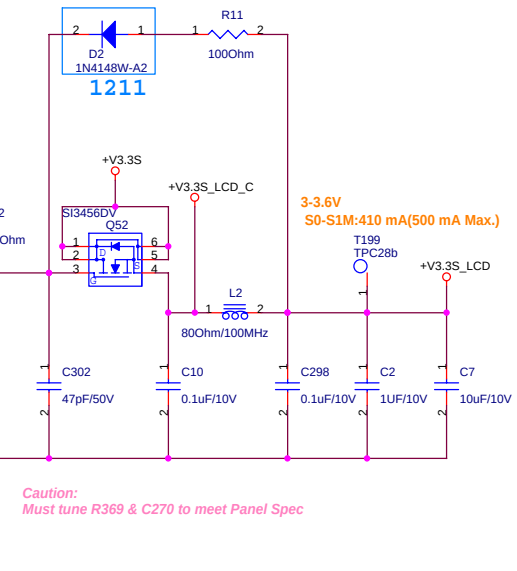
USB PORT 5 for WLAN

1219



LCD Power

SI3865: US\$0.22



Caution:
Must tune R369 & C270 to meet Panel Spec

LVDS & BACKLIGHT

Place Pi-Filter close to CRT
(≤ 200 mils)

Guarded by GND (Space ≥ 20 mils)

Length Matching (± 100 mils)

Guarded by GND (Space ≥ 20 mils)

Length Matching (± 100 mils)

Guarded by GND (Space ≥ 20 mils)

Length Matching (± 100 mils)

CRT
PIN

CRT CONNECTOR

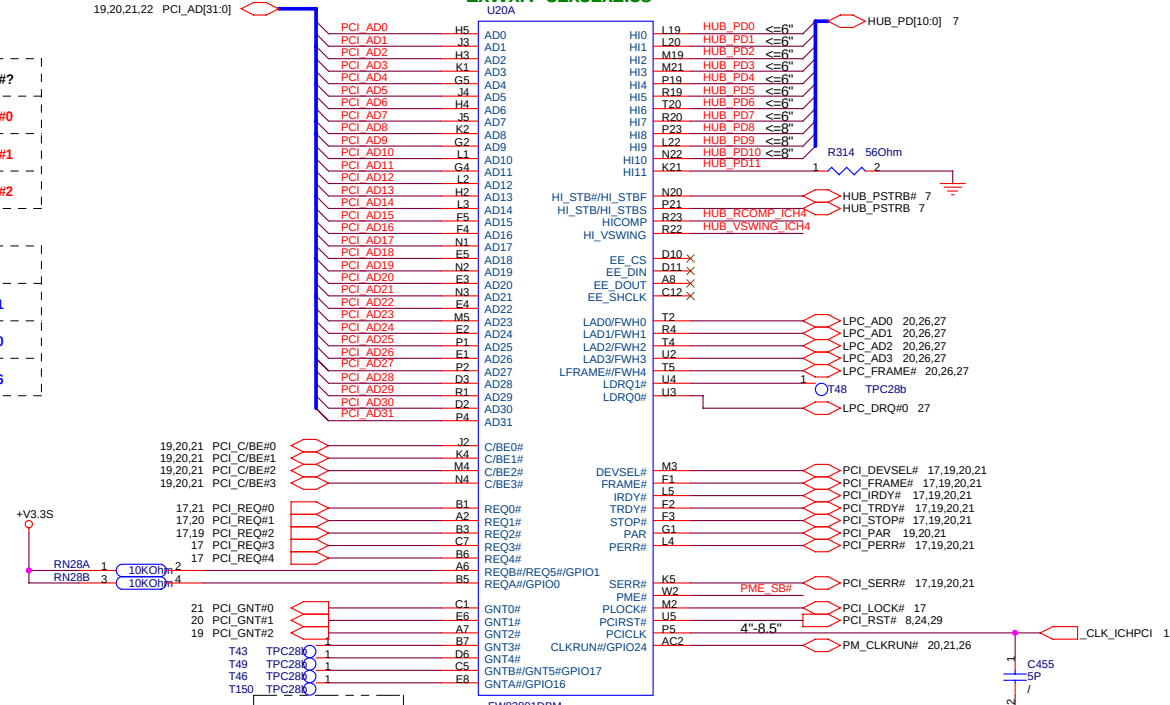
Use Daisy-Chain Topology

LxWxH=31x31x2.38

U20A

PCI_REQ#	PCI_REQ#?
CB&1394	PCI_REQ#0
MINIPCI	PCI_REQ#1
LAN	PCI_REQ#2

IDSEL	PCI_AD?
CB&1394	PCI_AD21
MINIPCI	PCI_AD20
LAN	PCI_AD16



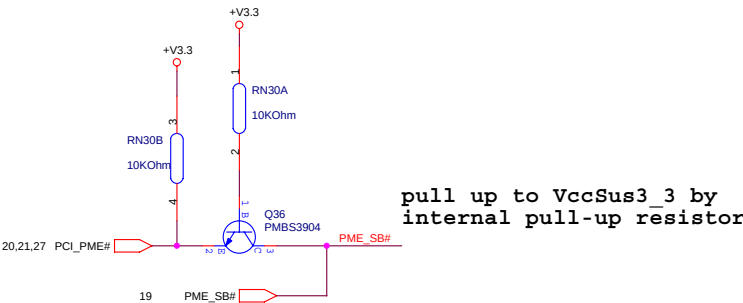
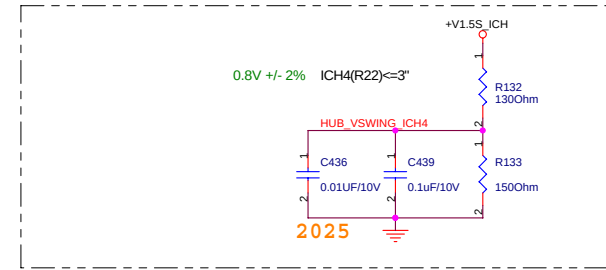
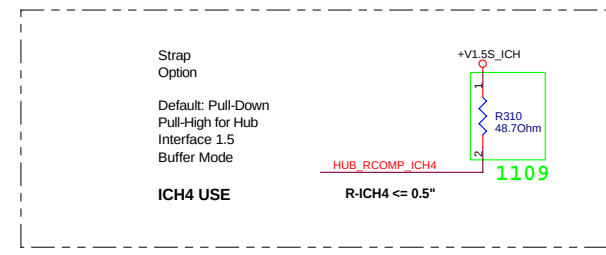
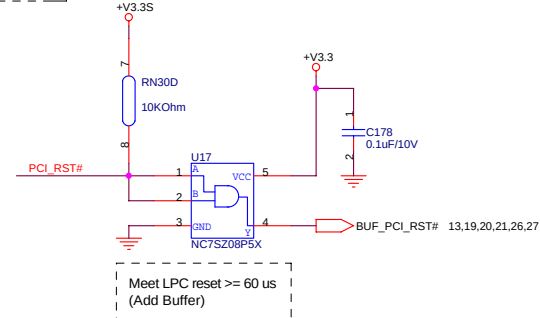
ICH4 pin E8

Strap Option

Default: Pull-High 20K

Pull-Down for BIOS

TOP-BLOCK SWAP



ICH4-M(HUBPCI)

USB SIGNALS
X Clock Signals
| USB+ - USB-| <= 150 mils
Pair Width/Space: 7/10 mils
Impedence: 90 ohm(differential)
Other Signals Space: >= 20 mils
Clock Signals Sapce:>= 50 mils

LxWxH=31x31x2.38

LxWxH=31x31x2.38

IDE I/F:
Width: 5 mils
Space: 7 mils
Length<= 8"
Match: <= 500 mils

IDE I/F:
Width: 5 mils
Space: 7 mils
Length<= 8"
Match: <= 500 mils

USBRBIAS:
WS: 5/5 mils
Length: <= 0.5"

GPIO[32:43] default:
Output High
ICH4 EDS R:1.0 page 10-94
10.10.8, 10.10.9

AC97 SDOUT & SYNC

Outer Layer: 5 mils
W/S= 1:1
ICH4-(T): 1"- 8"
R1-(T): 0.1"- 0.4"
AC97-R1: <=5.6"
R2-(T): 0.1"- 0.4"
AMC-R2: 0.9"-5.6"

1 to 8 inches
0.1 to 0.4 inches
0.9 to 5.6 inches

T type routing, place R at branch point.

Strap Option Default: Pull-Down 20K Pull-High for CPU SAFE_MODE

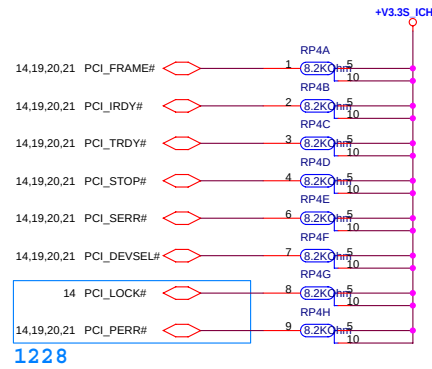
INTEL REQUEST

AC97 SDIN
ICH4 R2 AMC AC97
Outer Layer: 5 mils
W/S= 1:1
ICH4-R2: 0.9"- 13.6"
AC97-R2: 0.1"-0.4" (#0013)

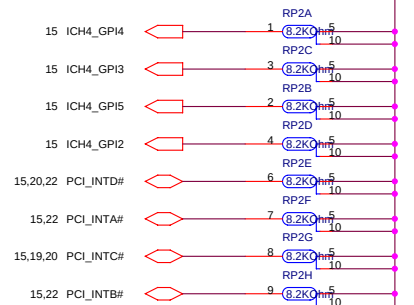
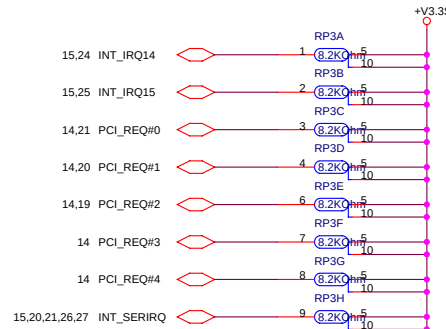
PID 0 ICH4	3	2.2KOhm	RN89B	PID_0	12
PID 1 ICH4	1	2.2KOhm	RN89A	PID_1	12
PID 2 ICH4	5	2.2KOhm	RN89C	PID_2	12
PID 3 ICH4	7	2.2KOhm	RN89D	PID_3	12

2021

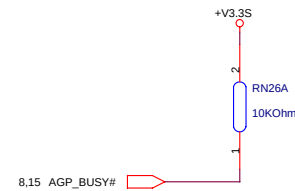
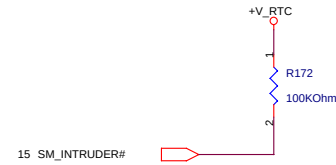
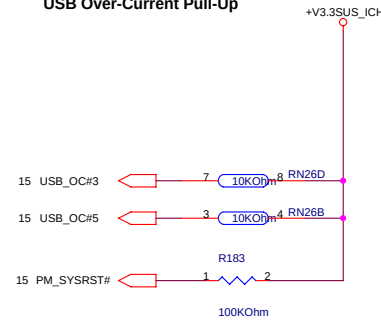
ICH4-M(H_U_IDE_PM)



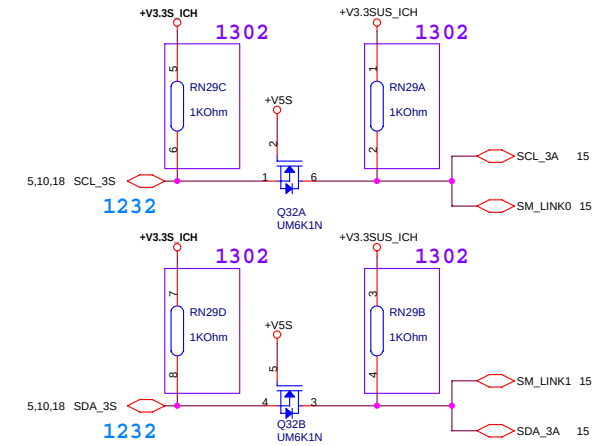
Can Swap



USB Over-Current Pull-Up

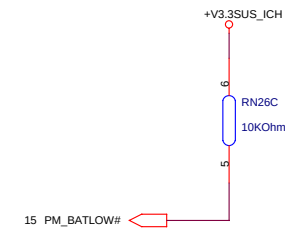


ICH4 SMLink & SMBus must be tied together



BATTERY LOW

ICH4

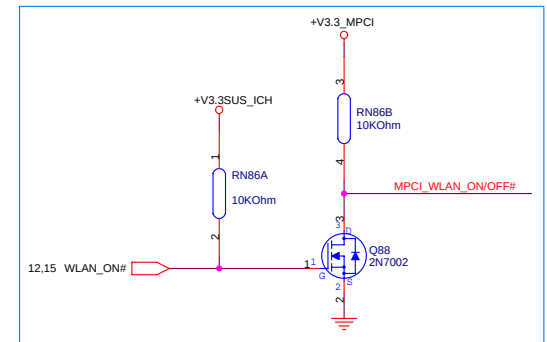
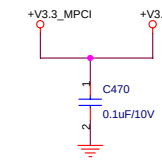
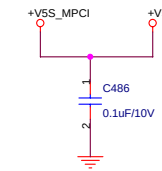
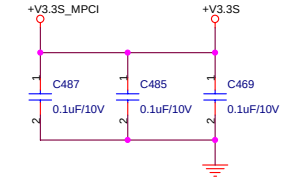
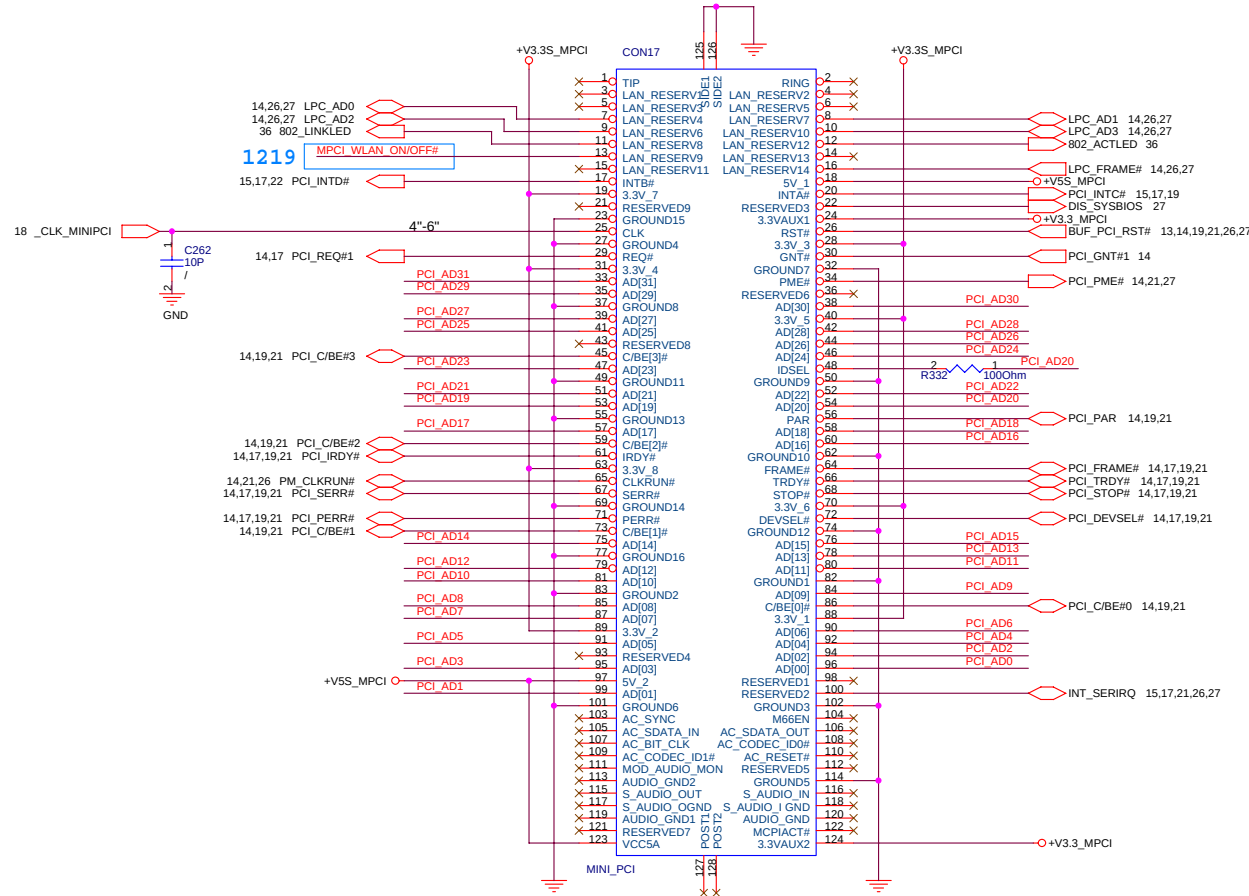


ICH4-M(PULLUP)



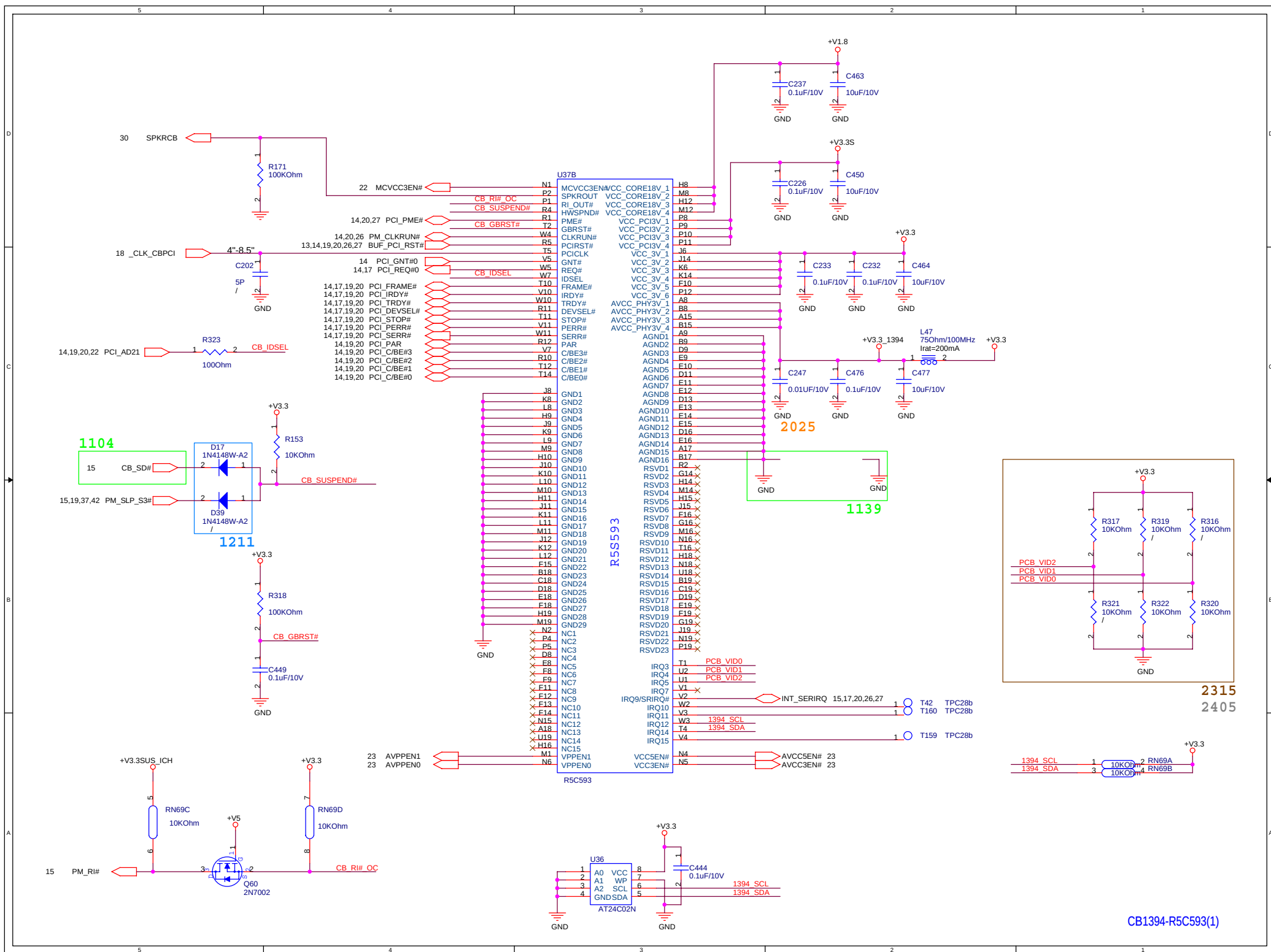
14.19.21.22 PCI_AD[31:0]

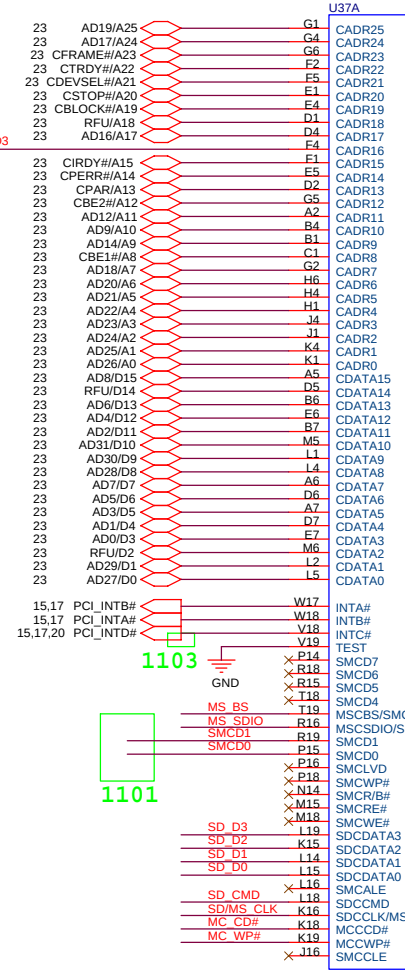
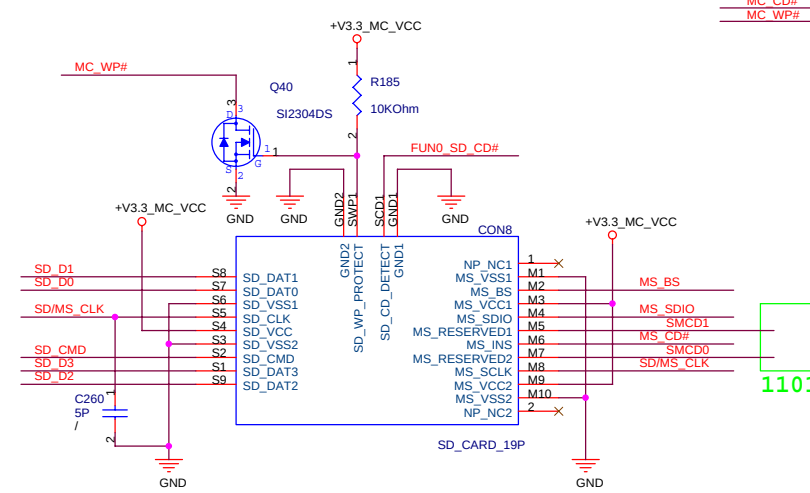
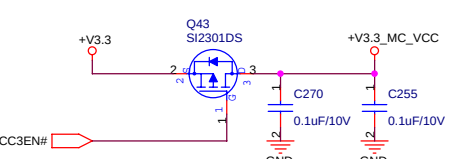
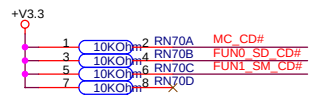
2208



1219

MINIPCI



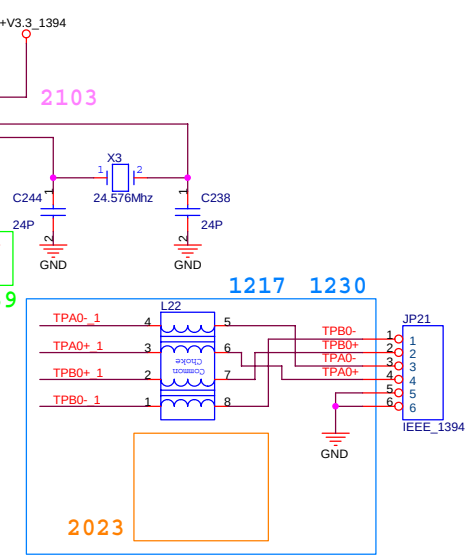
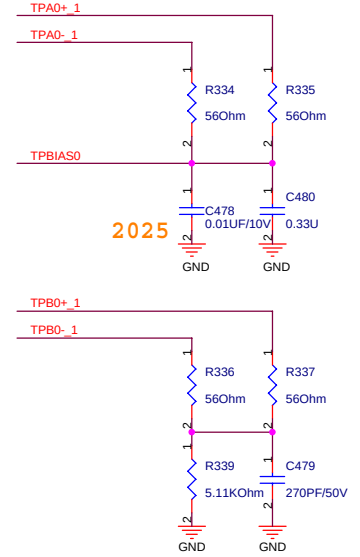
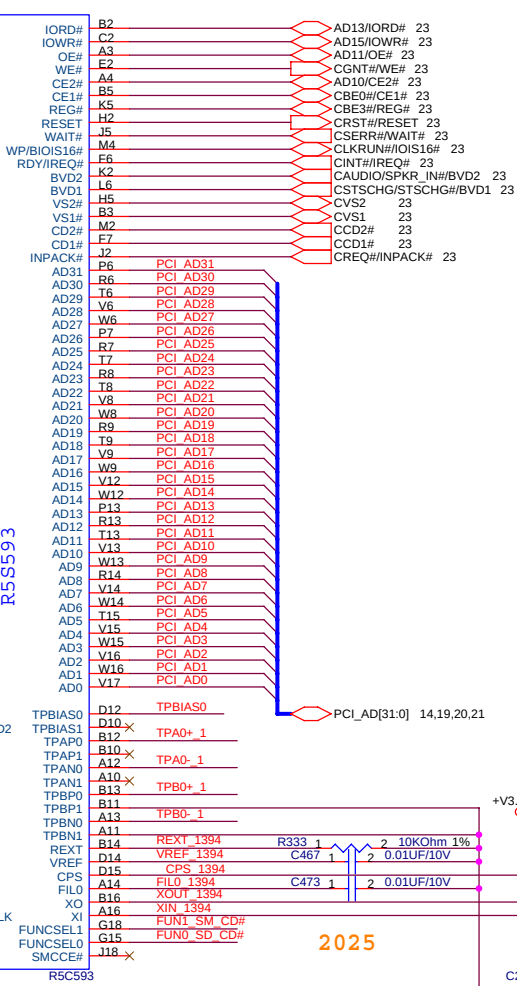
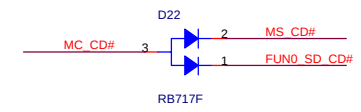


Memory Card Detect

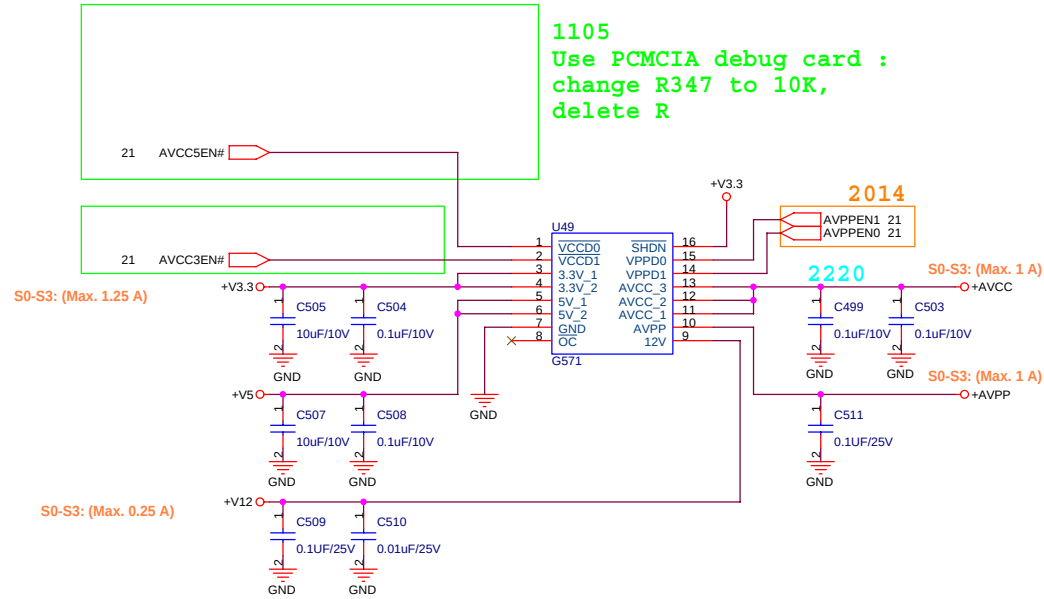
FUNSEL1	FUNSEL0	
0	0	Not Support
0	1	SmartMedia
1	0	MMC/SD
1	1	Memory Stick

MC_CD# : Memory Card Detect

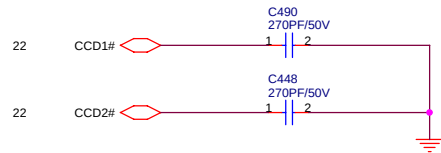
Turn-on voltage 0.37 V



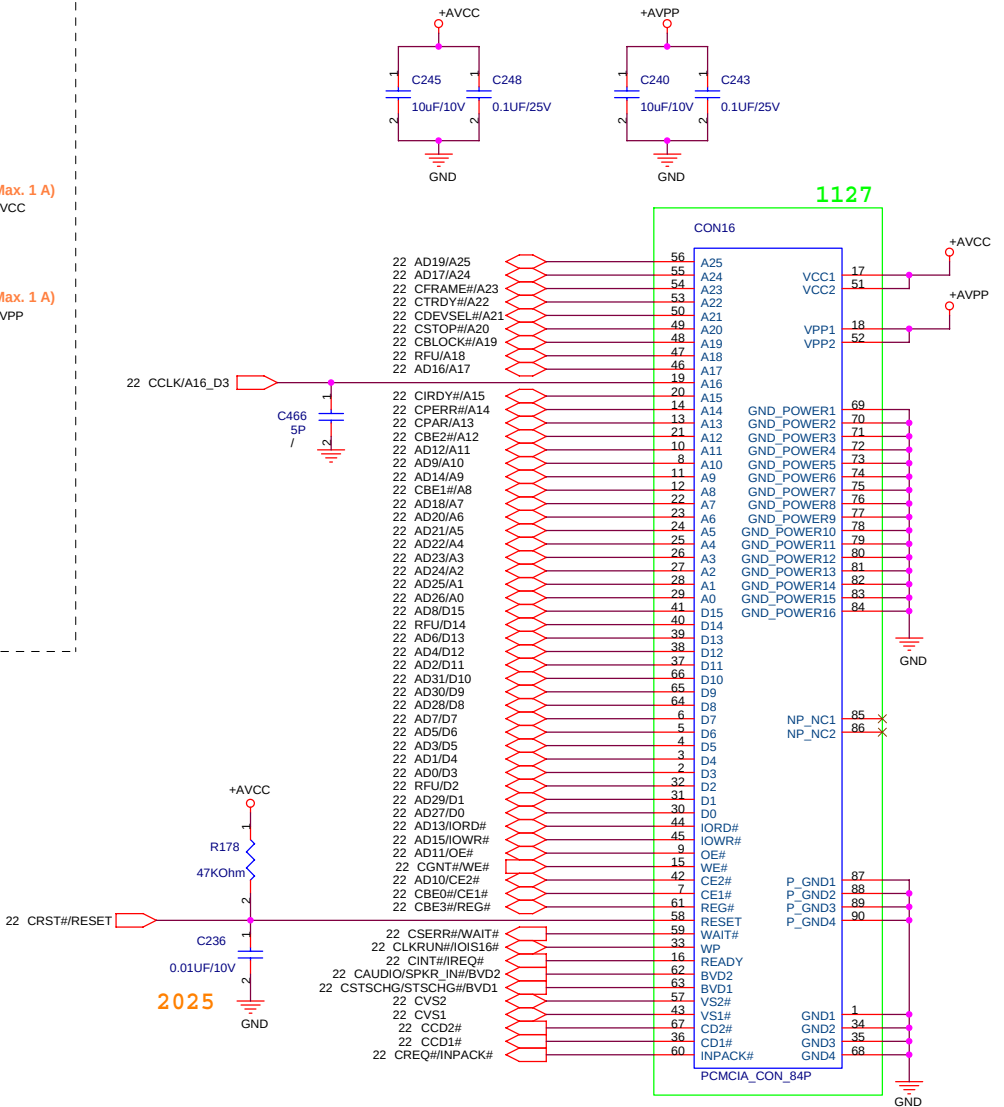
CB POWER SWITCH

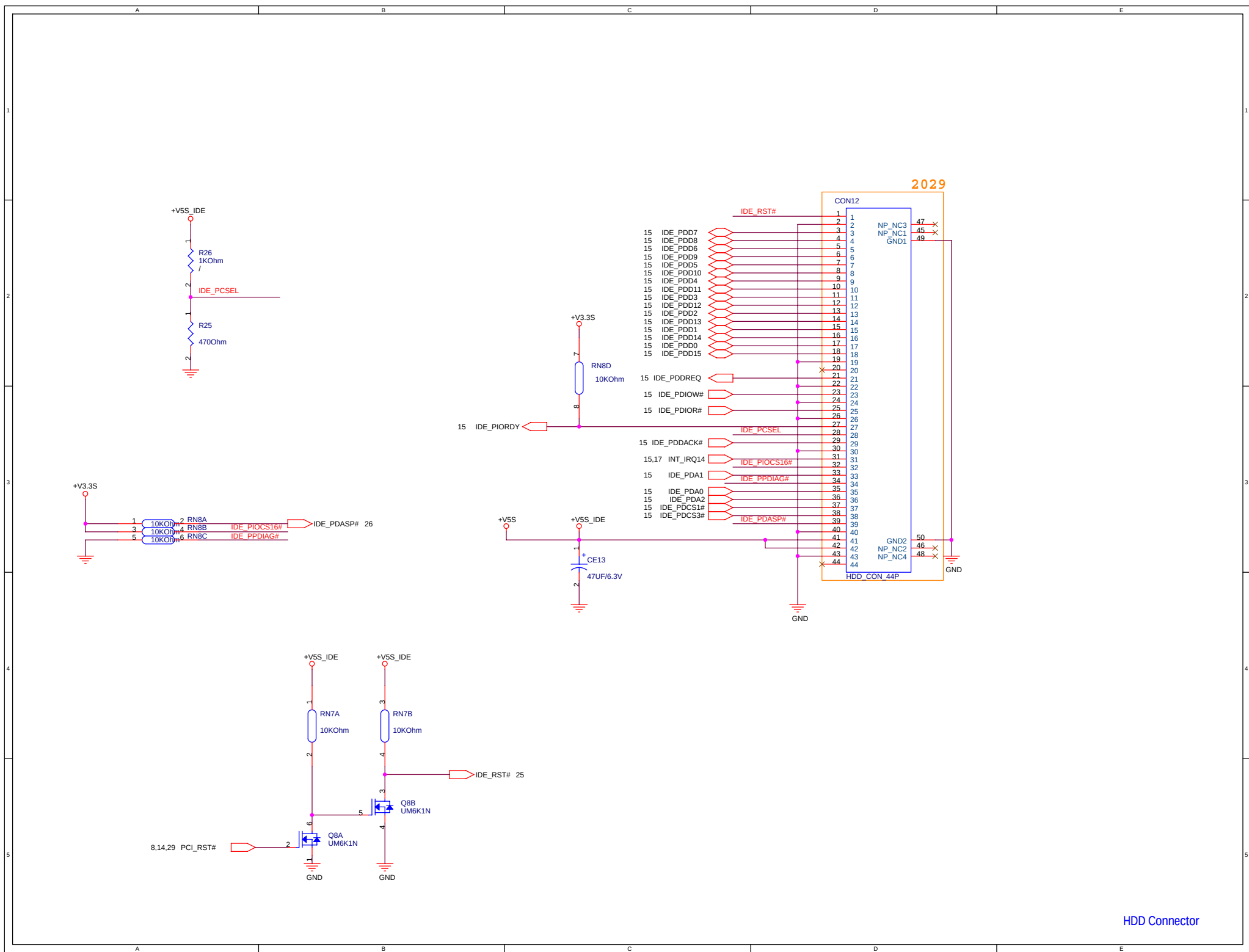


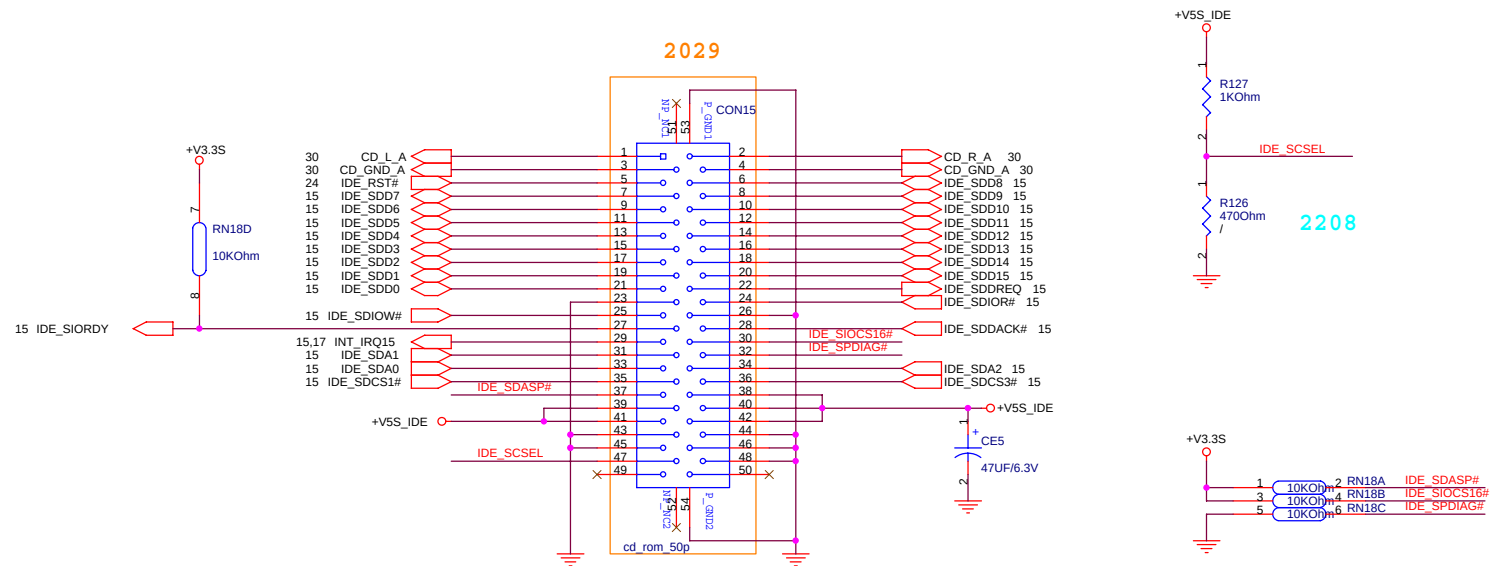
CB DE-BOUNCE



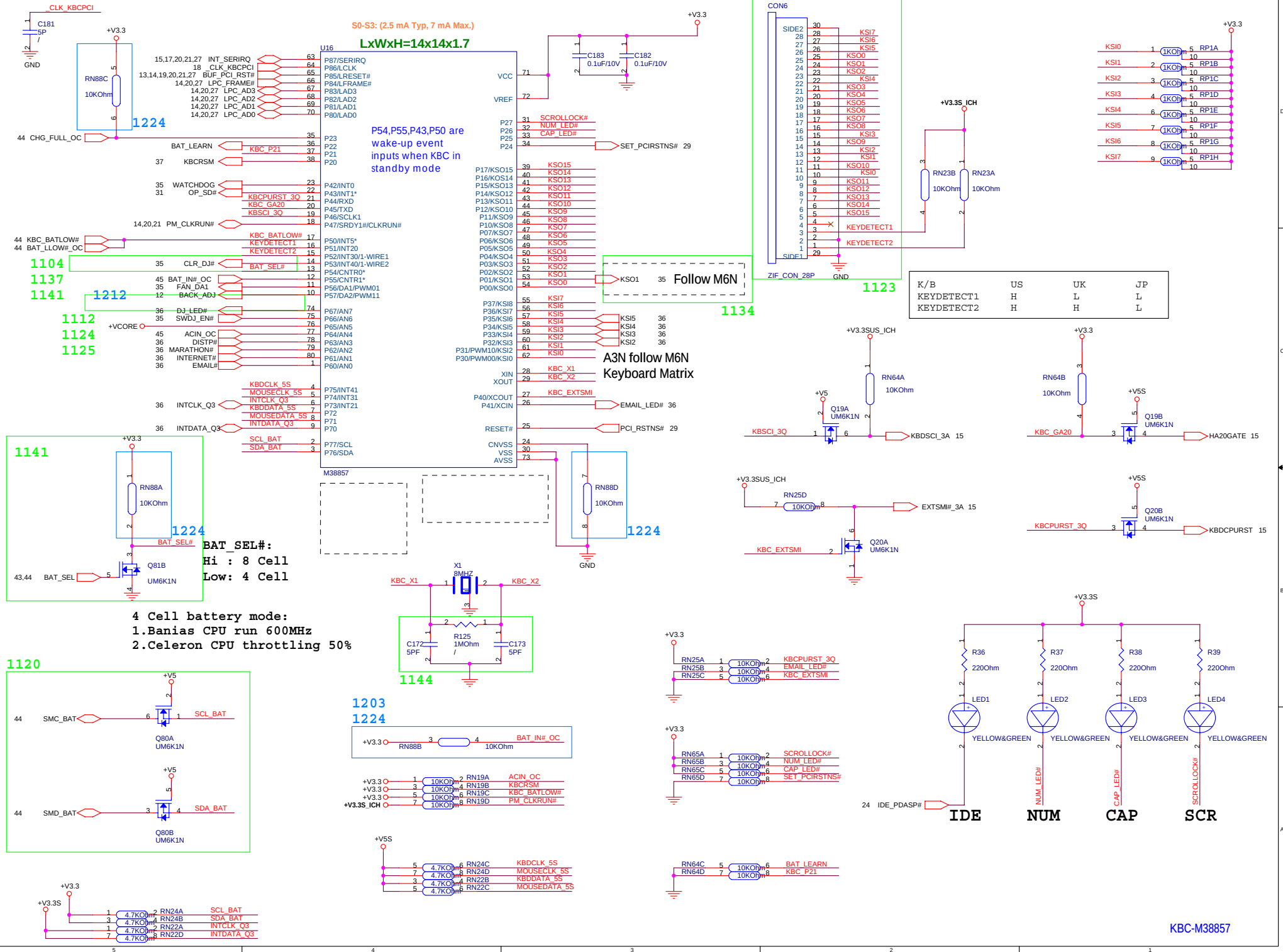
CB SOCKET



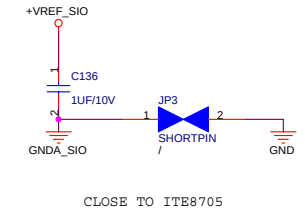
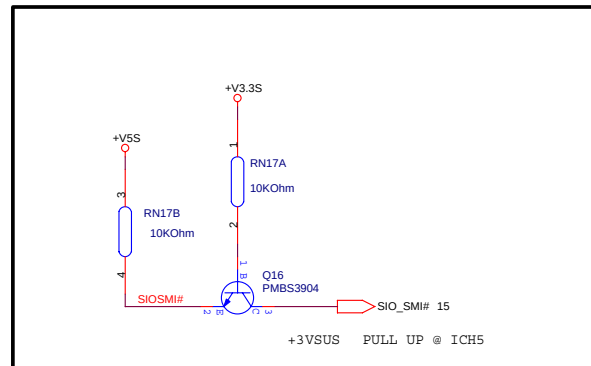
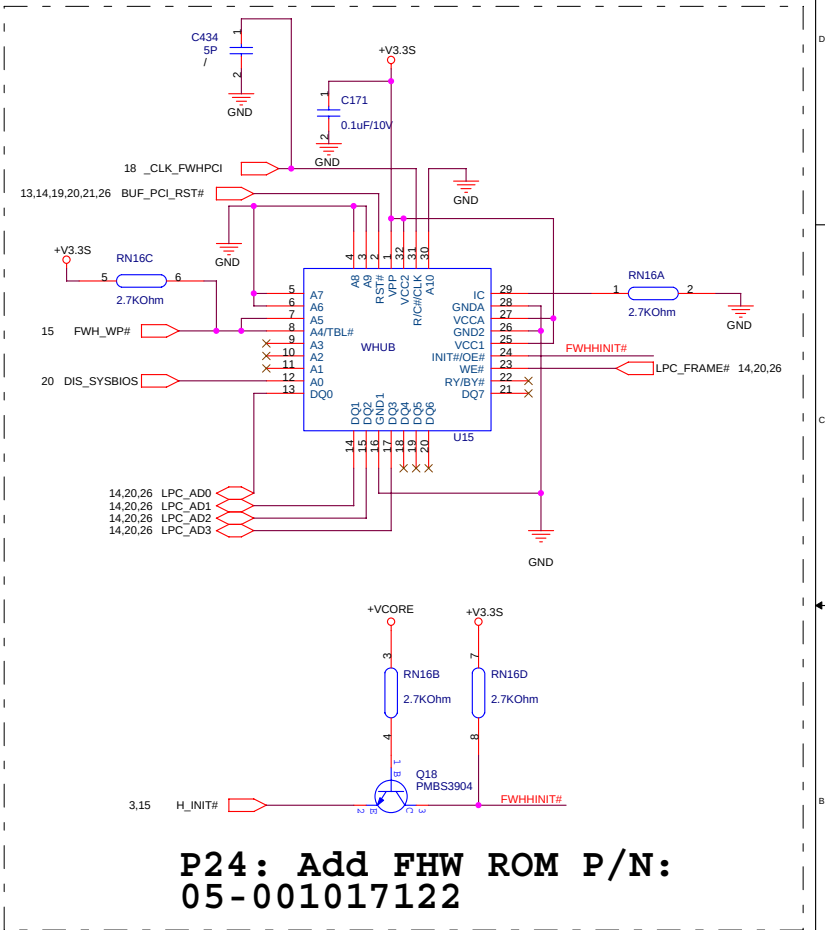
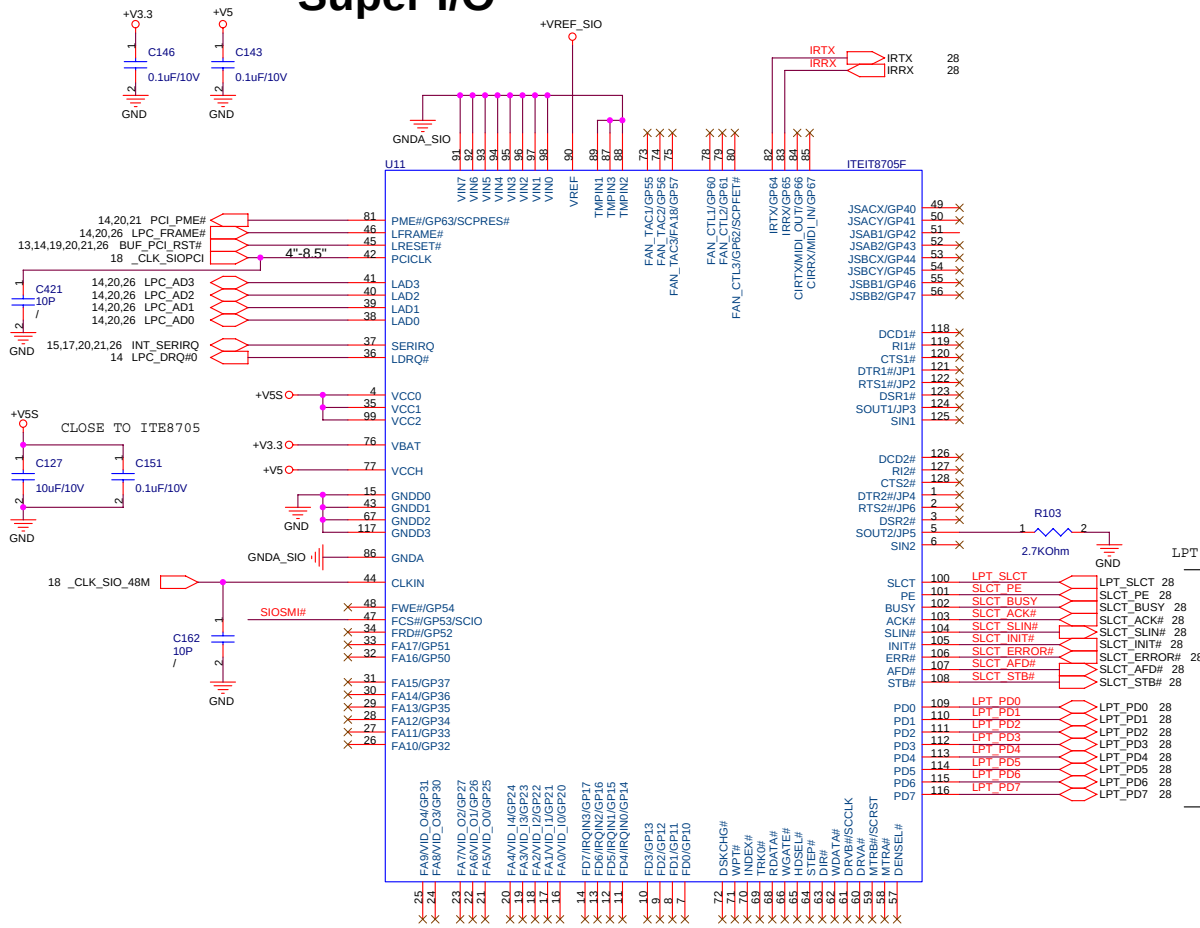




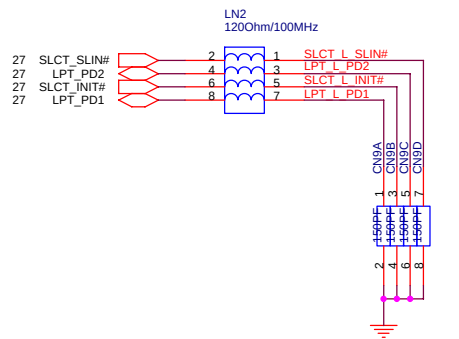
ODD Connector



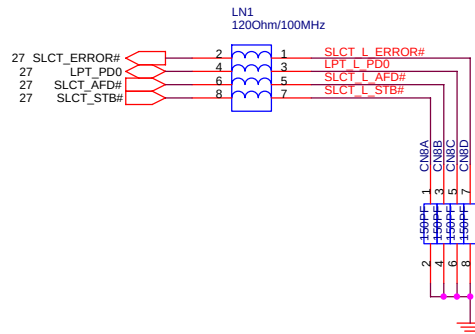
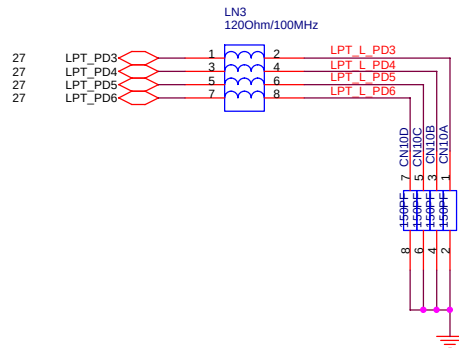
Super I/O 2208



SIO-ITE8705

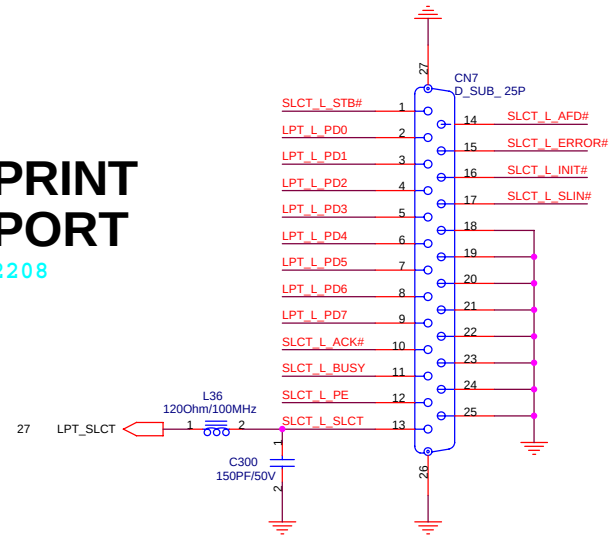


LAYOUT SWAP



PRINT PORT

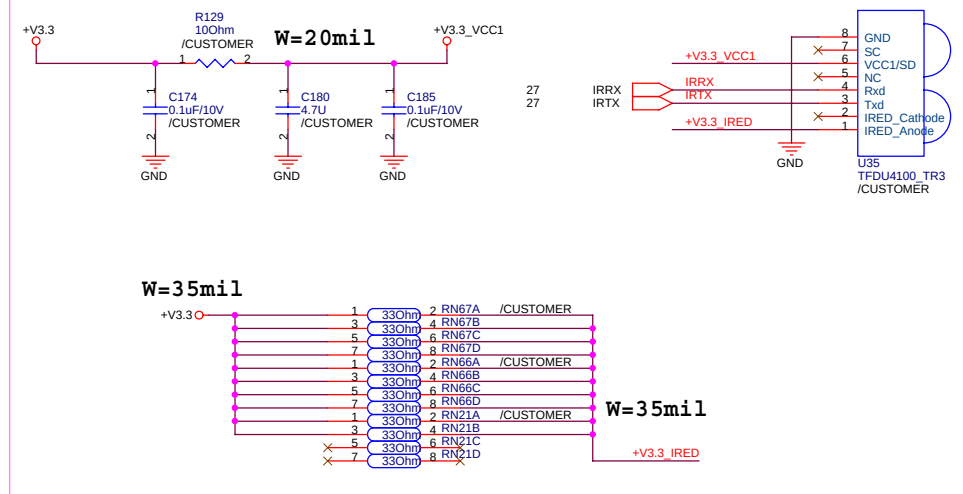
2208



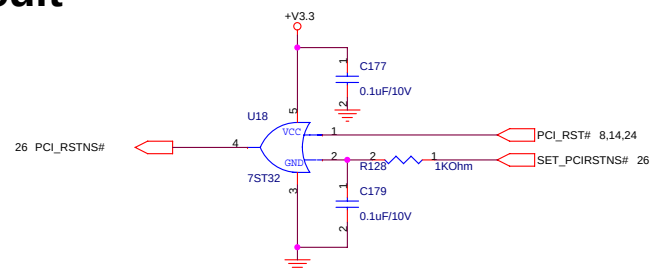
2309
2222 2408

2119

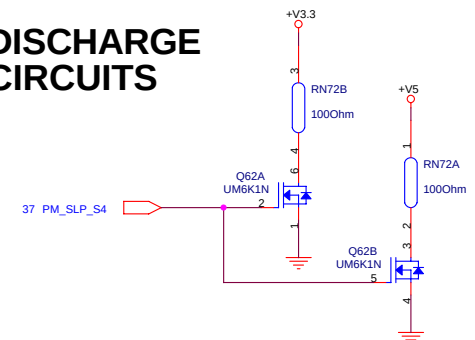
2208



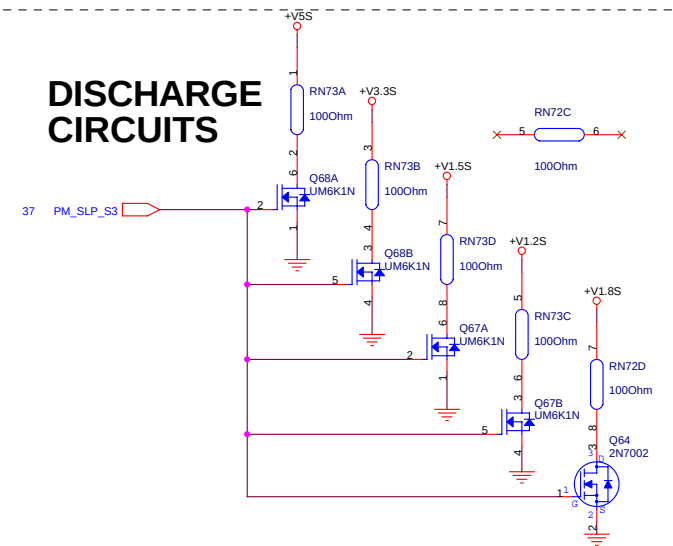
PCI_RSTNS# Gen Circuit



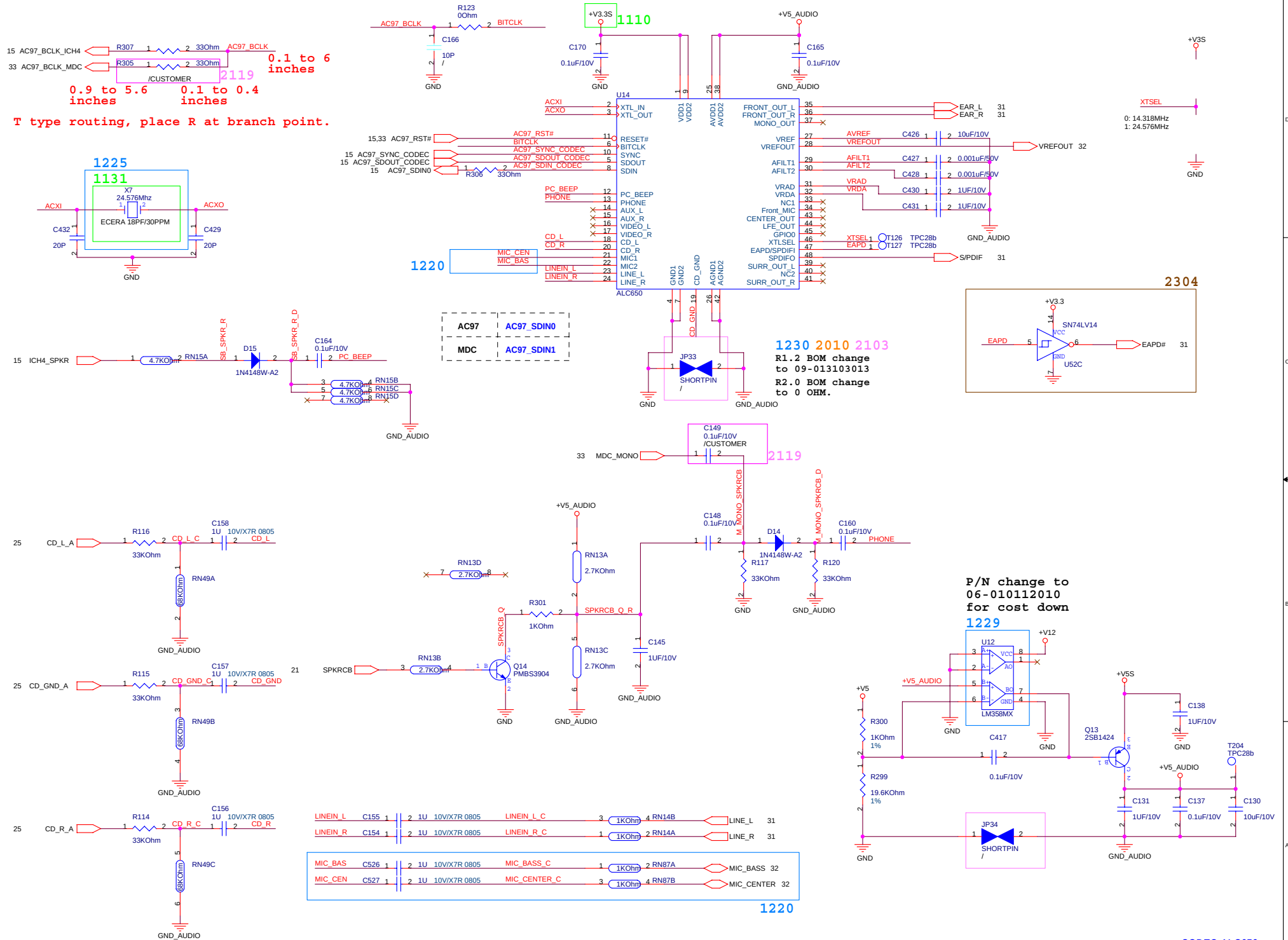
DISCHARGE CIRCUITS

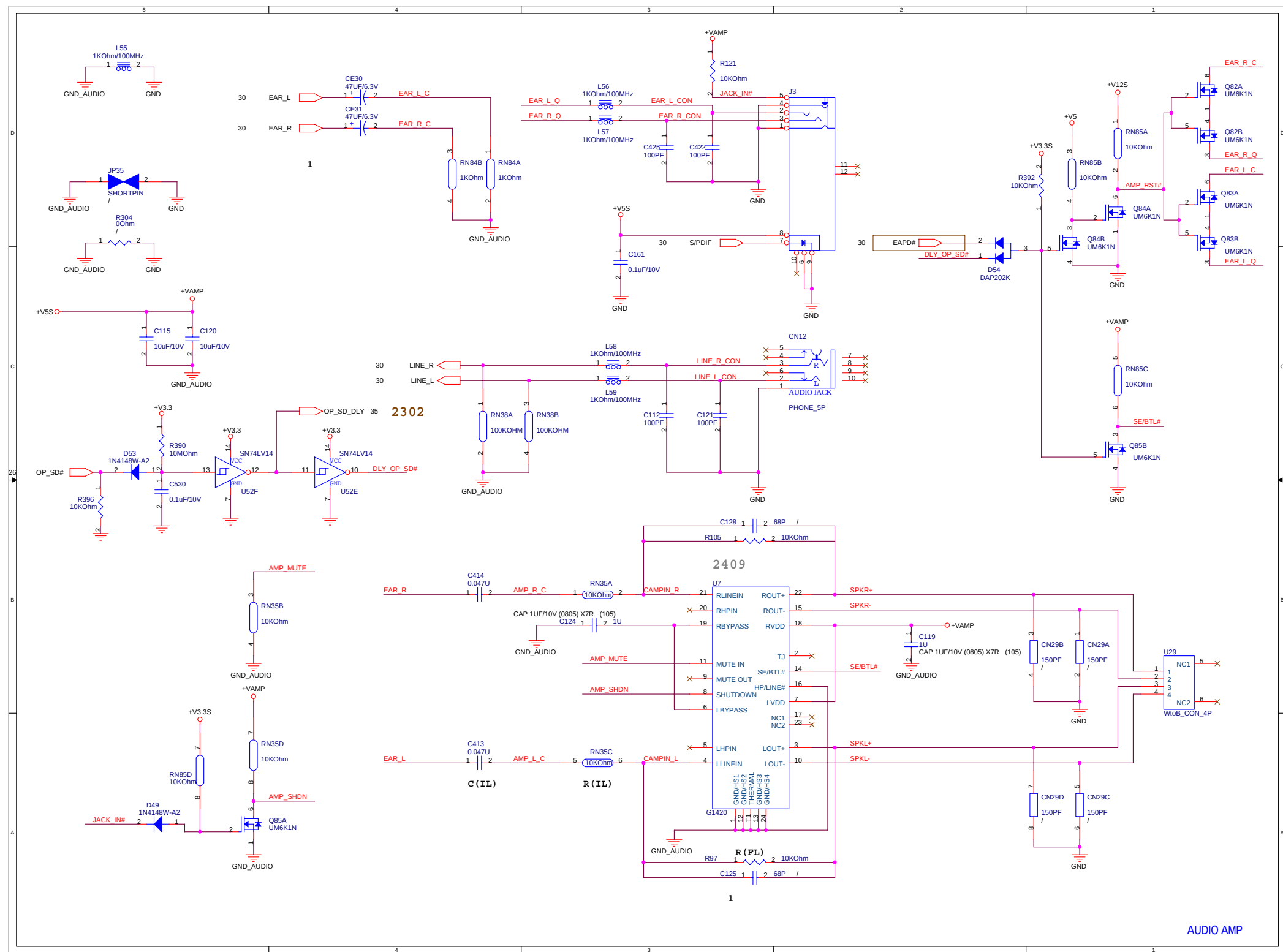


DISCHARGE CIRCUITS

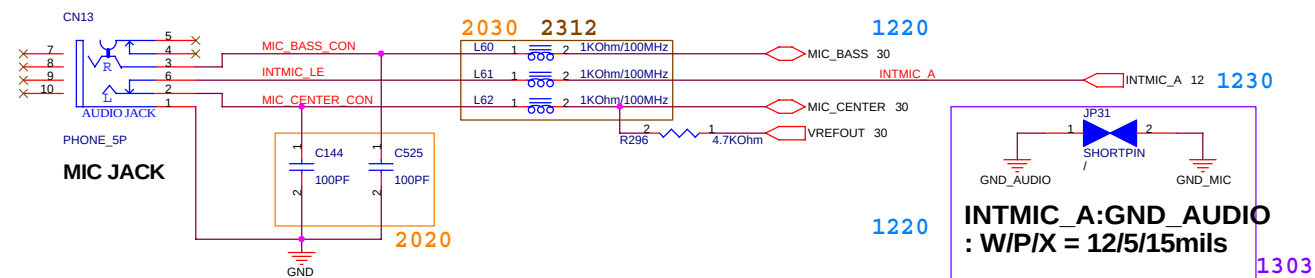


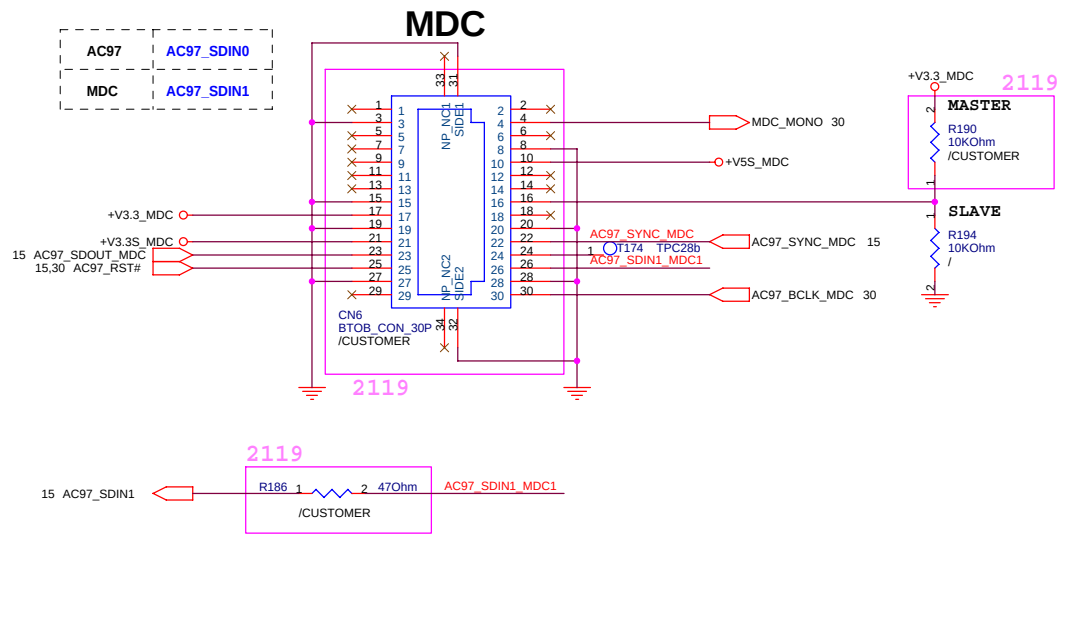
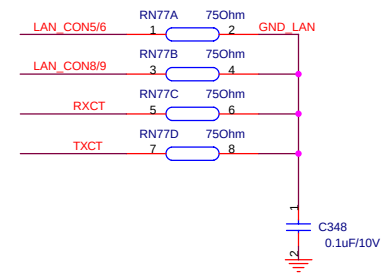
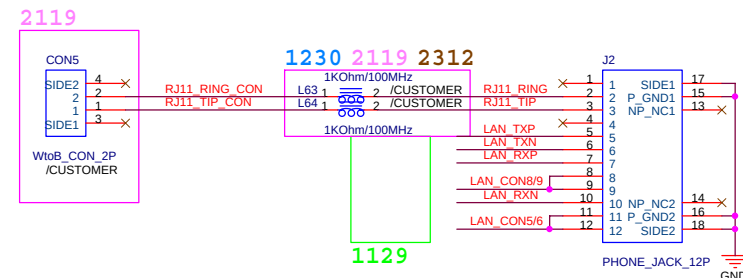
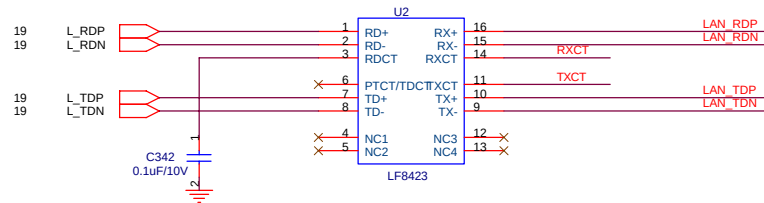
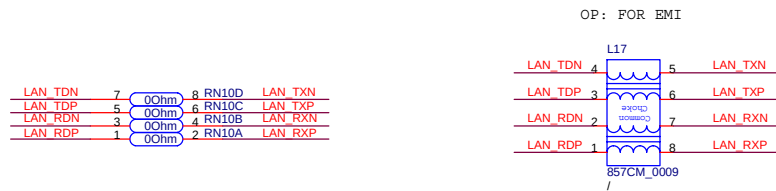
Discharge circuit





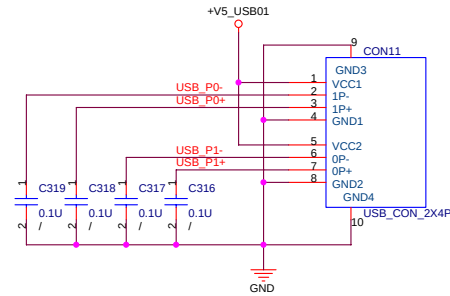
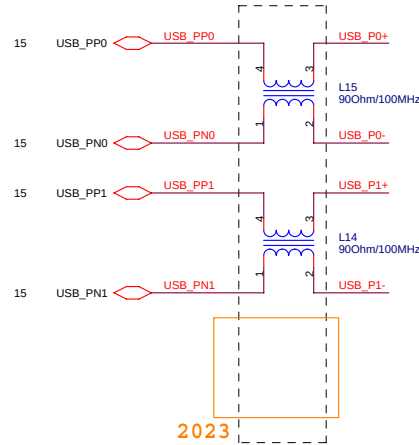
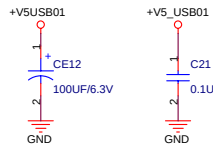
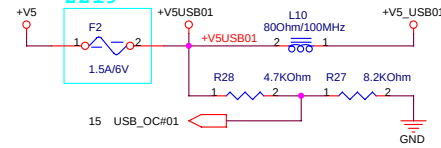
MIC JACK





VPLP R:2.2 use 07-014110900 (2nd source is 07-014110201)

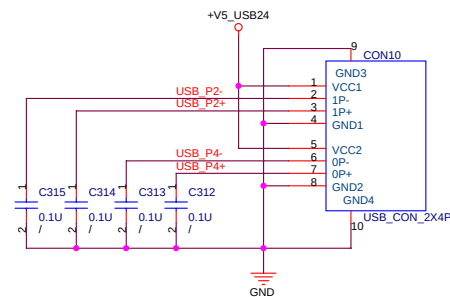
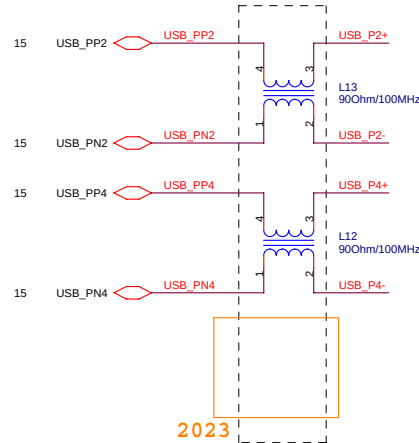
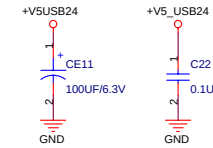
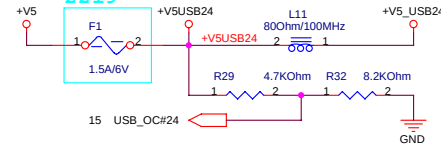
2219



USB PORT 0 & PORT 1

VPLP R:2.2 use 07-014110900 (2nd source is 07-014110201)

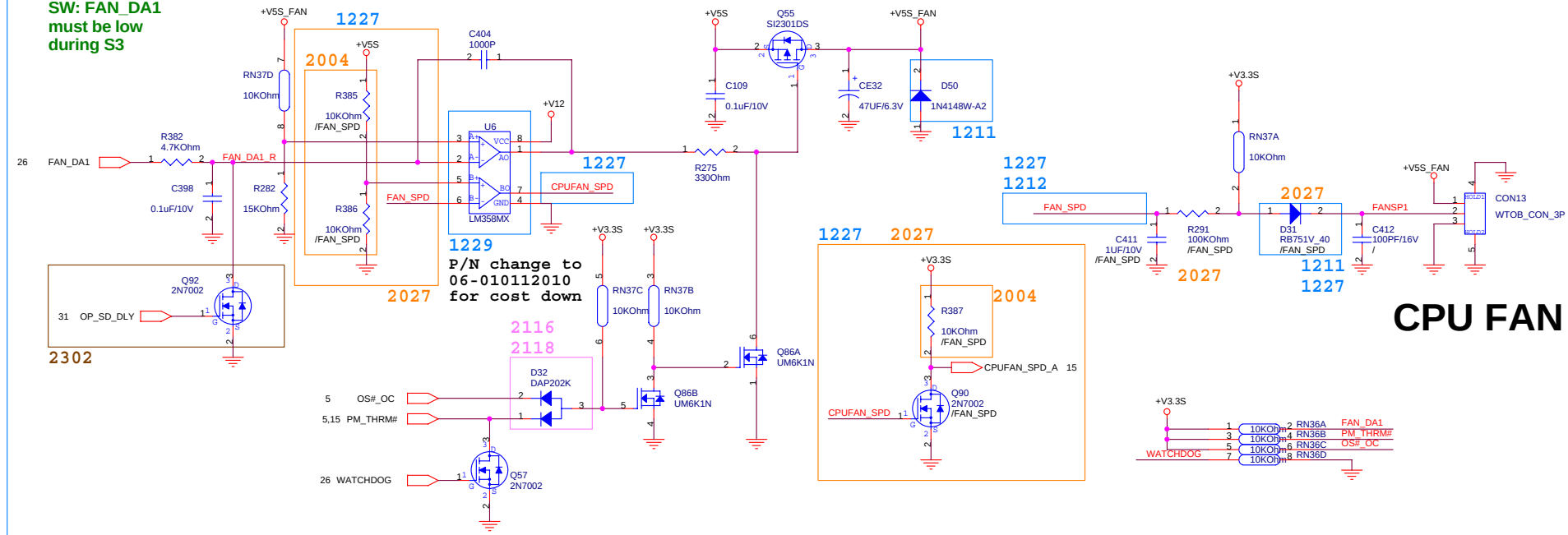
2219



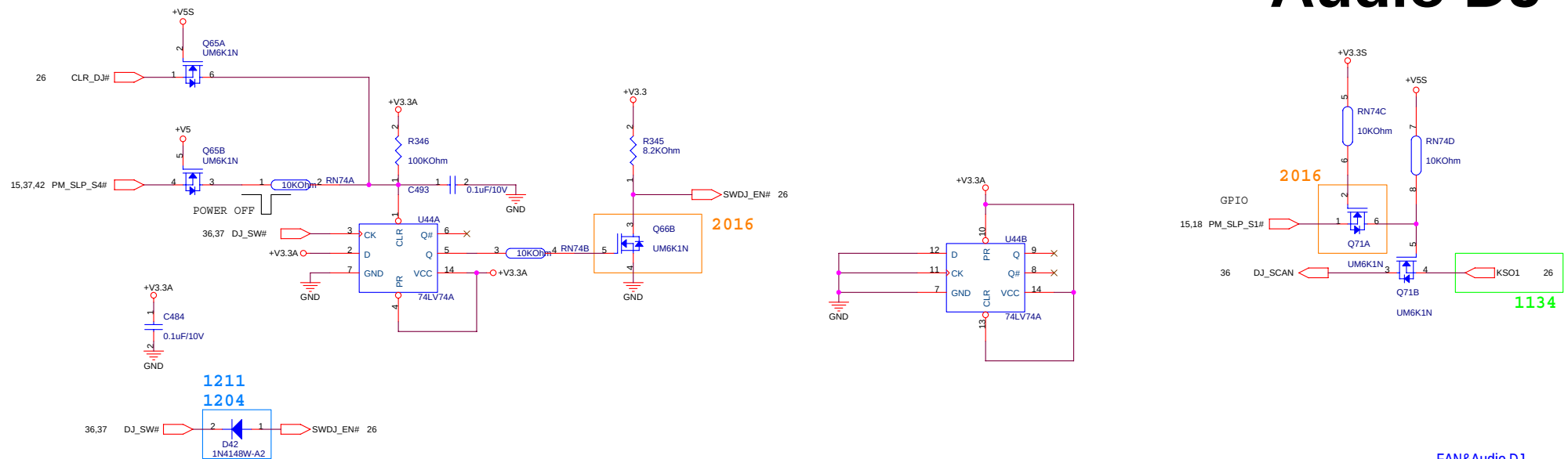
USB PORT 2 & PORT 4

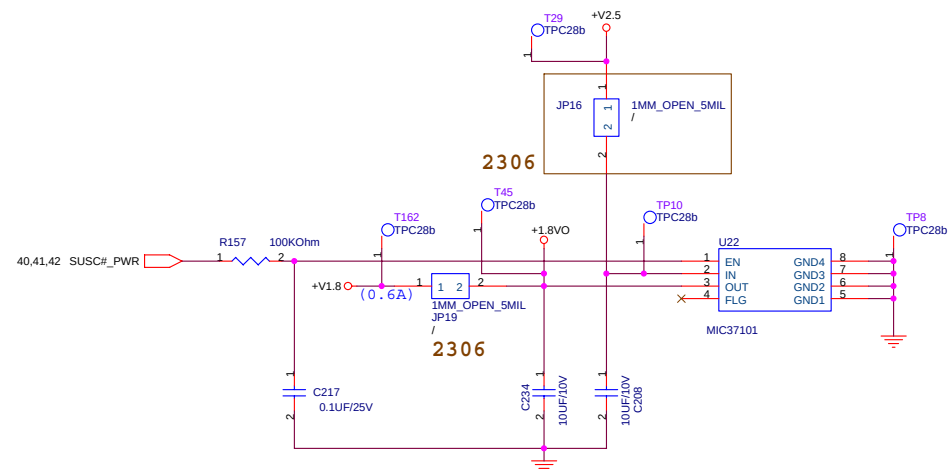
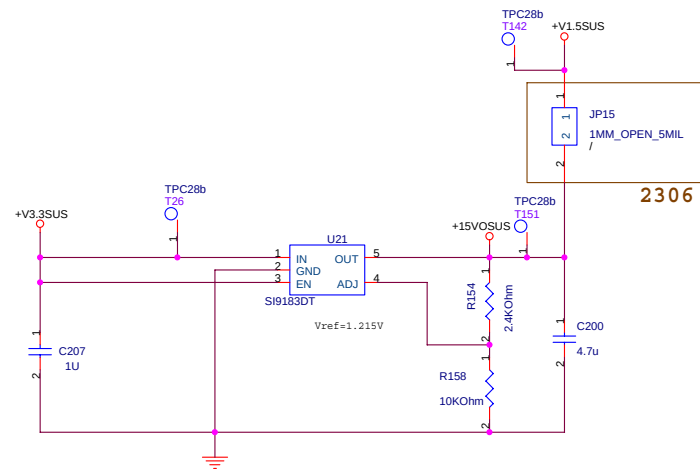
Fan Speed Control

SW: FAN_DA1
must be low
during S3

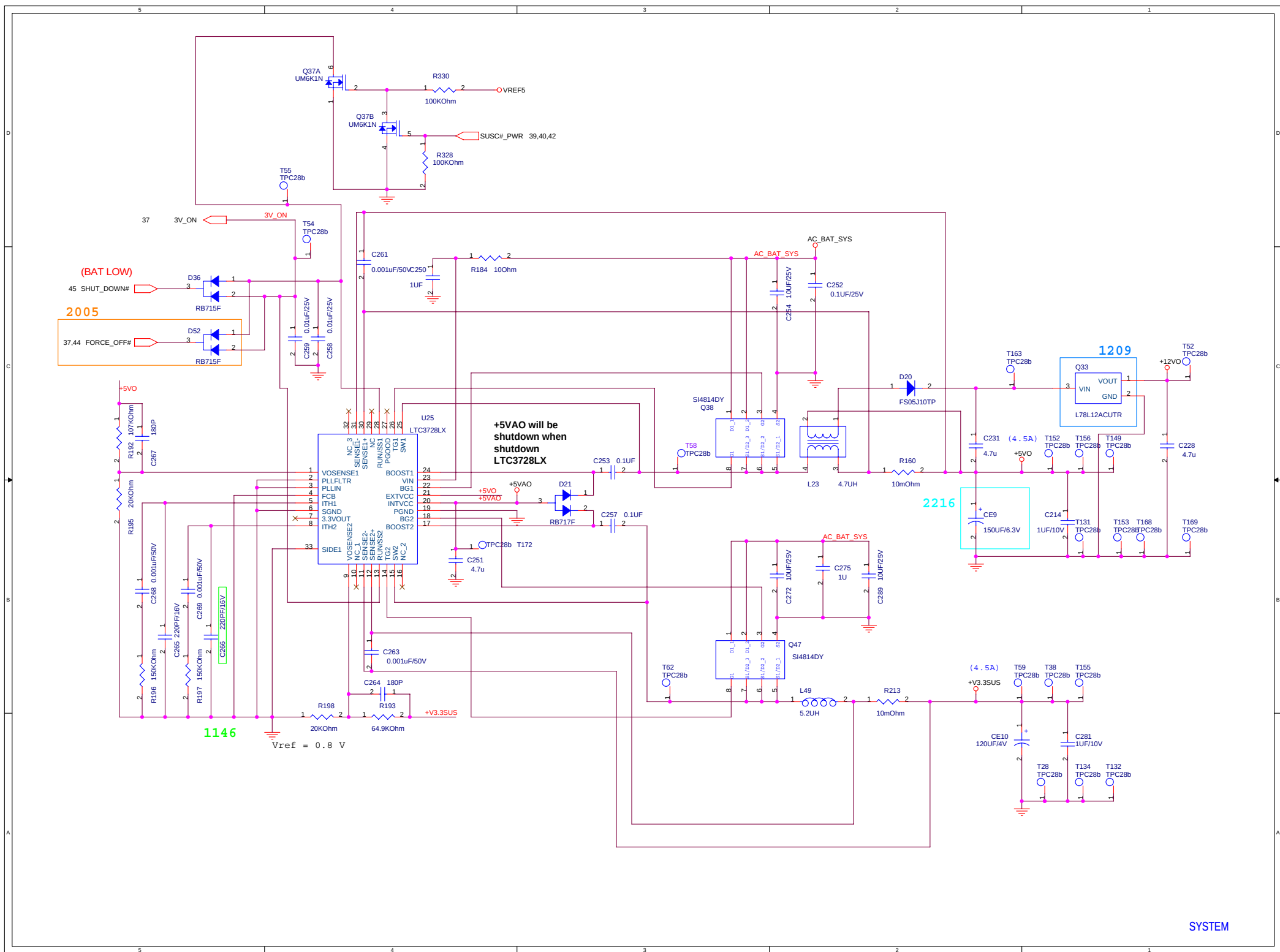


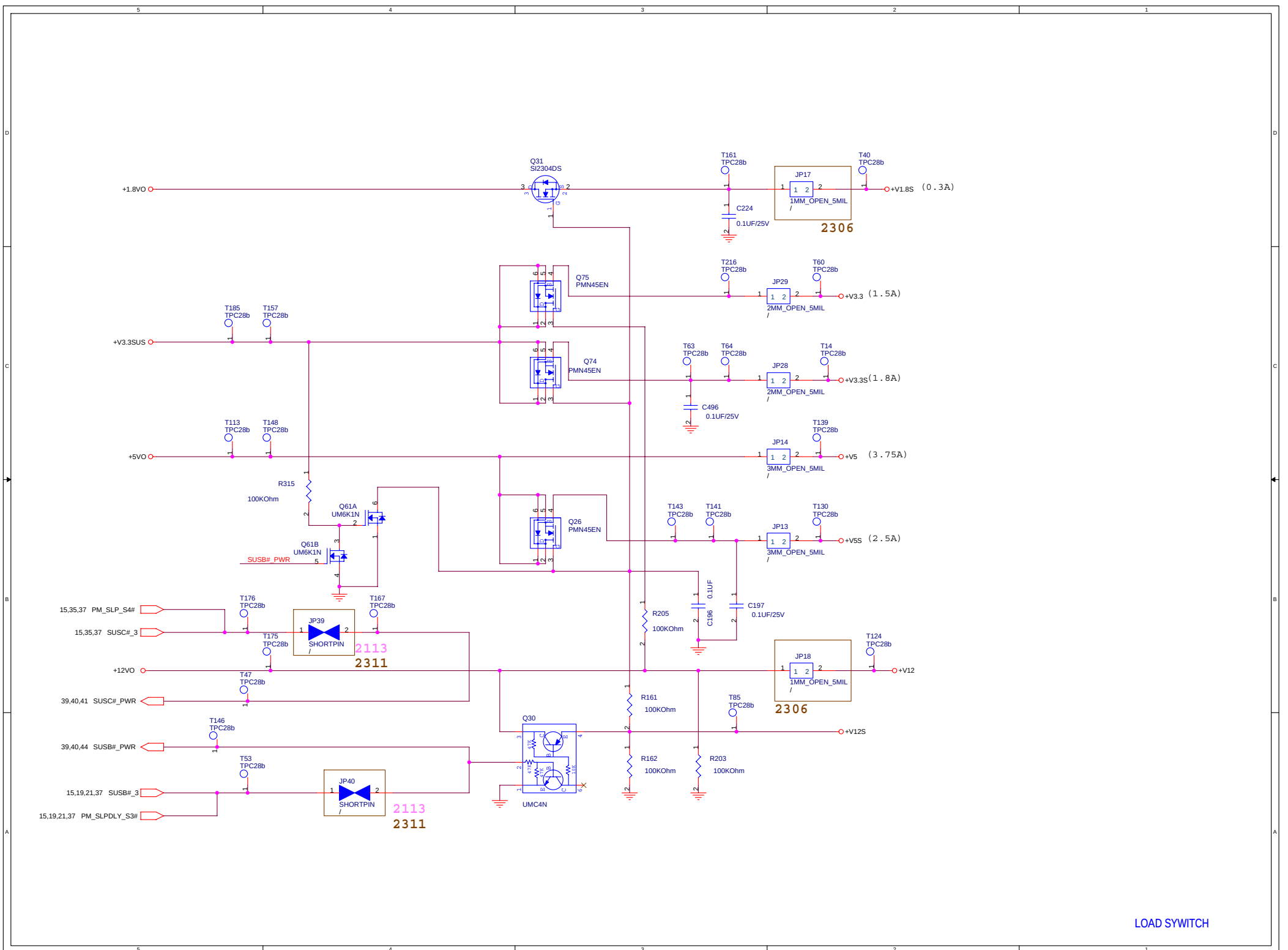
Audio DJ

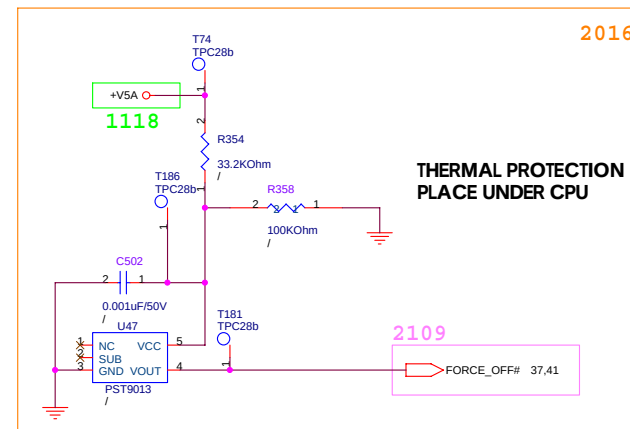
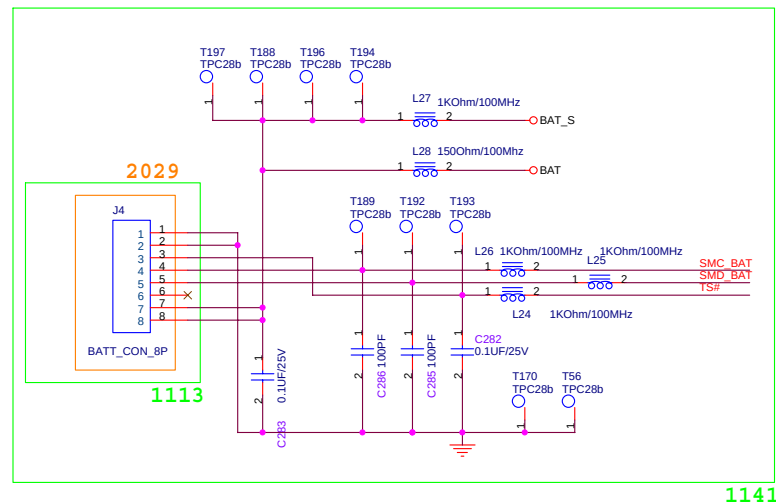
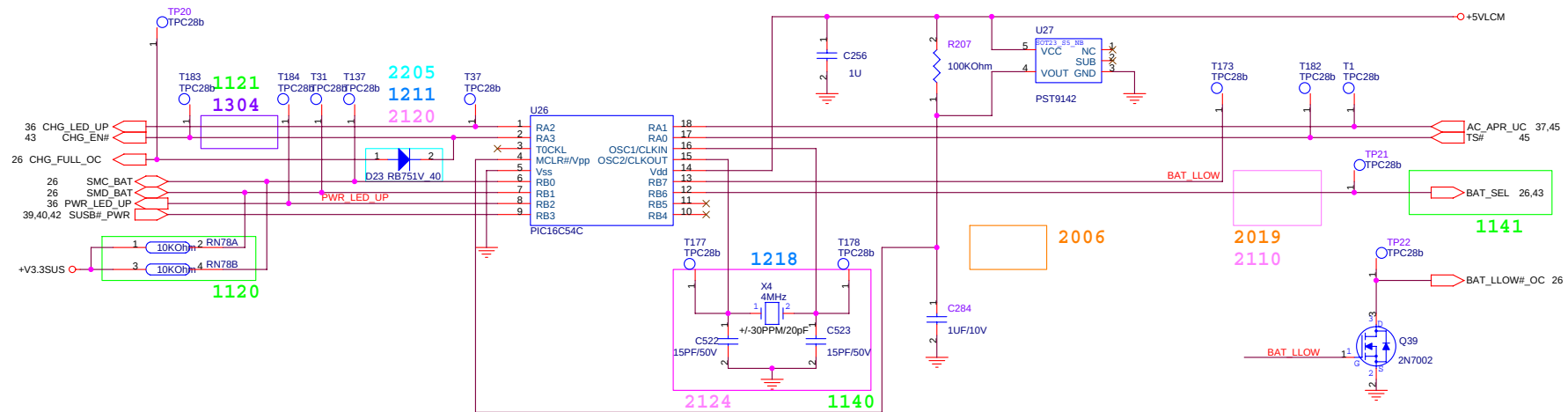


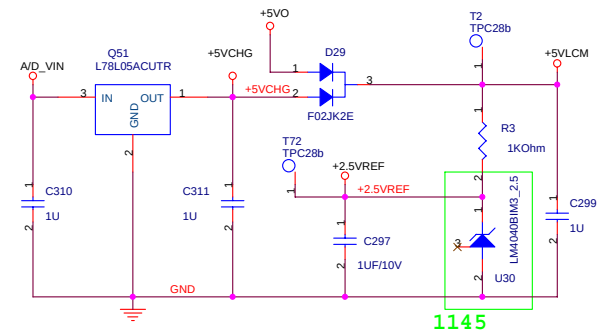
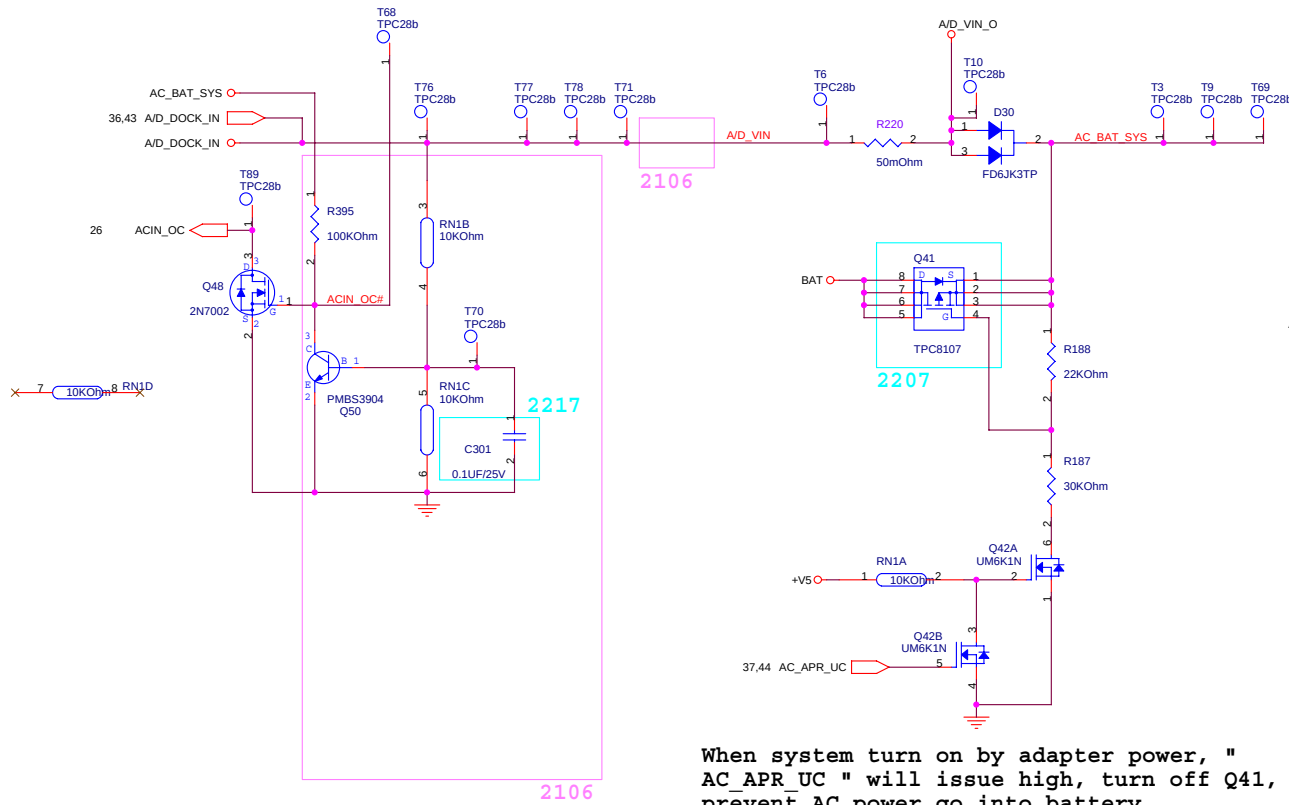
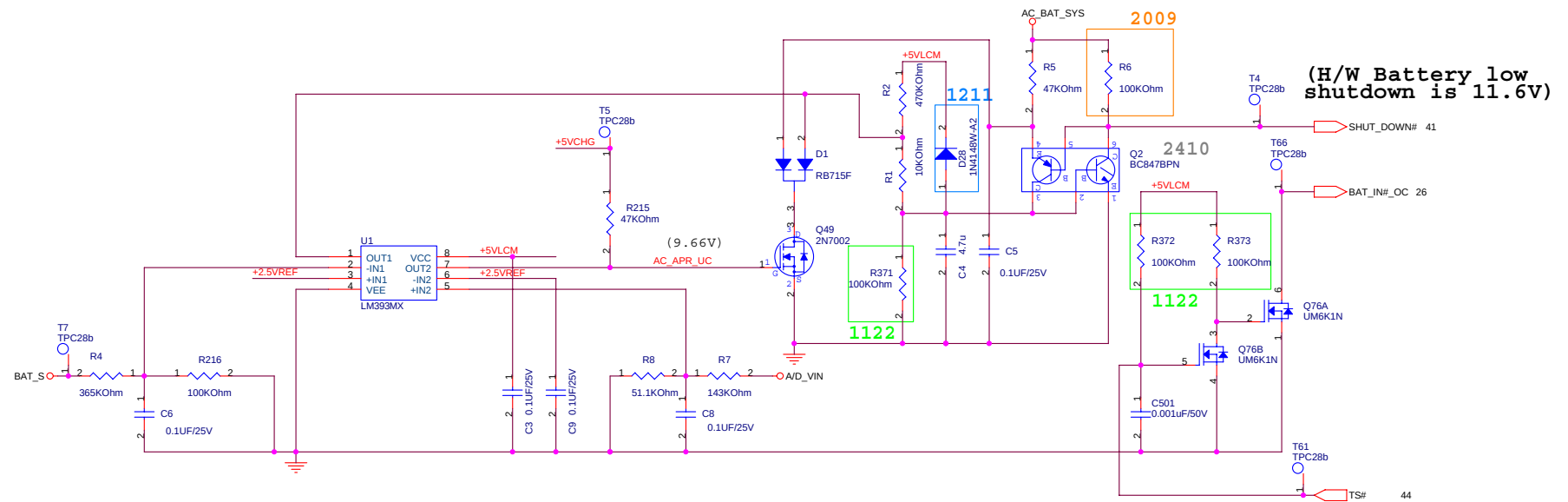


1.25V&1.8V

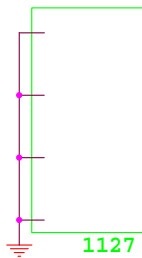




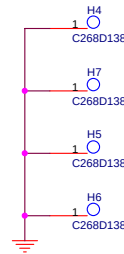
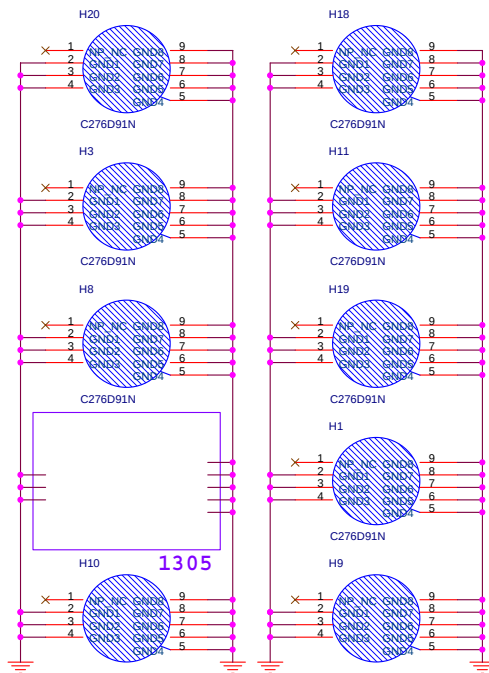
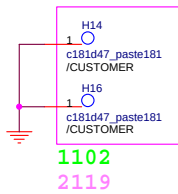




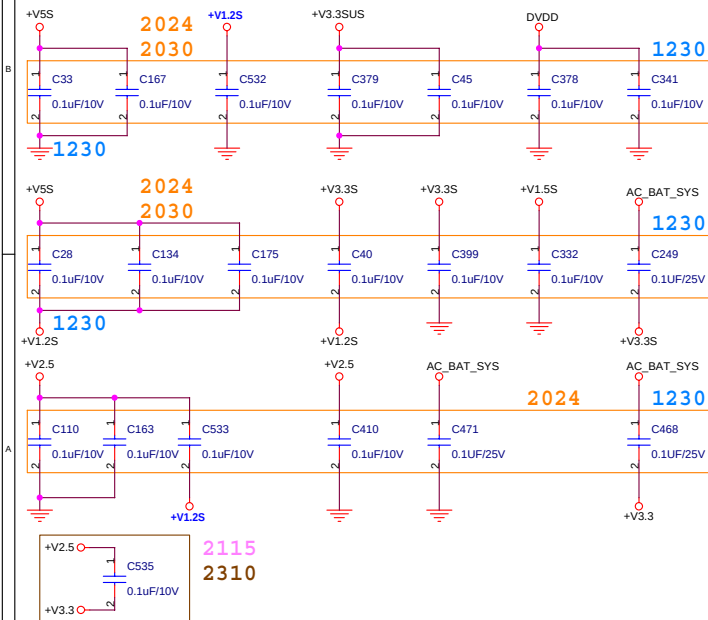
SCREW HOLE



For MDC module



EMI



Short Pad location:

Page37: JP32

Page39: JP4,5,15,16,19,26,27

Page40: JP6,7,9,22,24

Page42: JP13,14,17,18,28,29

Don't Short Pad: JP26

JP4:+V2.5 --Power In

JP5:+V1.25S

JP6:+V2.5

JP7:+V1.2S

JP9:+VCCP

JP13:+V5S

JP14:+V5

JP15:+V1.5SUS

JP16:+V2.5 --Power IN

JP17:+V1.8S

JP18:+V12

JP19:+V1.8

JP22:+V1.5S

JP24: VREF5-->+V5A

JP26:+3VALWAYS T-->+V3.3A (open)

JP27:+3VALWAYS M-->+V3.3A (short)

JP28:+V3.3S

JP29:+V3.3

JP32:KBCRSM

PCB STACK-UP

PCB THICKNESS: 1.6 mm

- L1 TOP
- L2 VCC
- L3 GND
- L4 BOT

IMPEDENCE

Single-Ended

27.4 OHM WIDTH

TOP/BOT 20 mils

37.5 OHM WIDTH

TOP/BOT 12 mils

42 OHM WIDTH

TOP/BOT 10 mils

55 OHM WIDTH

TOP/BOT 5 mils

Differential

70 OHM WIDTH/SPACE

TOP/BOT 9 mils/ 5 mils

90 OHM WIDTH/SPACE

TOP/BOT 7 mils/ 10 mils

100 OHM WIDTH/SPACE

TOP/BOT 5 mils/ 7 mils

PCI INTERFACE

PCI_REQ#

CB&1394 PCI_REQ#0

MINIPCI PCI_REQ#1

LAN PCI_REQ#2

PCI_GNT#

CB&1394 PCI_REQ#0

MINIPCI PCI_REQ#1

LAN PCI_REQ#2

IDSEL

CB&1394 PCI_AD21

MINIPCI PCI_AD20

LAN PCI_AD16

PCI_INT#

CB&1394 PCI_INTB/A/D#

MINIPCI PCI_INTC/D#

LAN PCI_INTC#

POWER INTERFACE

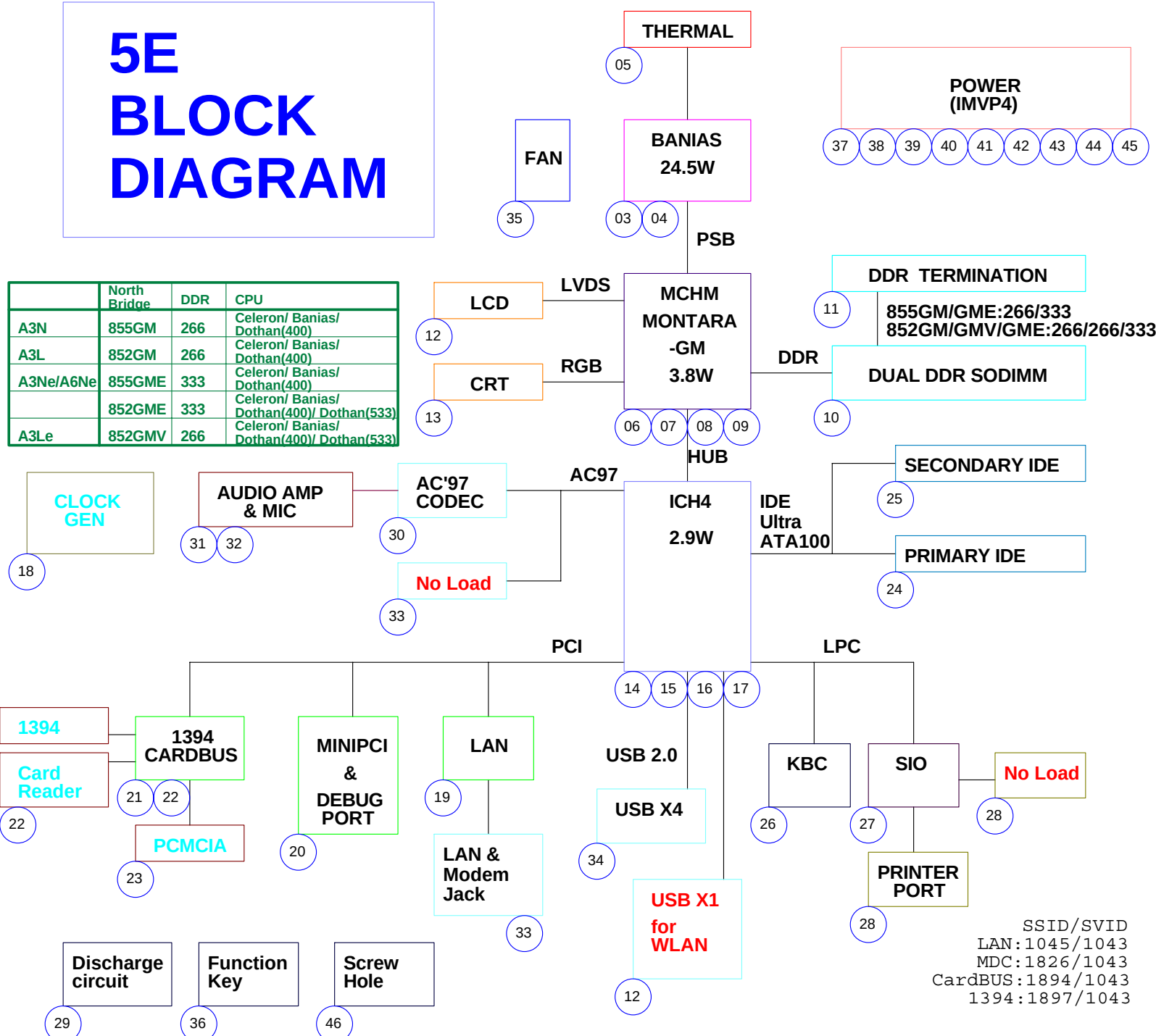
SIGNALS	TYPE	POWER
CLK_EN#	I	+V3.3S_CLK
PM_PSI#	O	+VCCP
VR_VID[5:0]	O	+VCCP
CPU_VRON	O	+V3.3SUS
VRM_PWRGD	I	+V3.3S
PM_STPCPU#	O	+V3.3S
CHG_LED	I	+5VLCM
RST_BTN#	O	OD
OTP_RESET#	I	+V5
SHUT_DOWN#	I	AC_BAT_SYS
+5VLCM	PWR	+V5
PM_SLPDLY_S3#	O	+V3.3
PM_SLP_S4#	O	+V3.3SUS
BAT_LEARN	I	+V3.3
BAT_LLOW#_OC	I	+V3.3
BAT_IN#_OC	I	+V3.3
ACIN_OC	I	+V3.3
CHG_FULL_OC	I	+V3.3
PM DPRSLPVR	O	+V3.3S
AC_APR_UC	I	+V5A
+V5A	PWR	VREF5
3V_ON	O	OD
AC_BAT_SYS	PWR	DC
A/D_DOCK_IN	PWR	DC
SMC_BAT	IO	+V3.3
SMD_BAT	IO	+V3.3

POWER PLANE

POWER	VOLTAGE	CURRENT
+VCORE	1.46V	25A
+VCCP	1.05V	2.4A(Max),1A(Real)
+V1.2S	1.2V	2.5A
+V1.25S	1.25V	0.5A
+V1.5S	1.5V	1.32A
+V1.5SUS	1.5V	64 mA
+V1.8	1.8V	0.14A
+V1.8S	1.8V	0.3 A
+V2.5	2.5V	6.68A
+V3.3S	3.3V	1.732A
+V3.3	3.3V	1.515A
+V3.3SUS	3.3V	14 mA
+V5S	5V	2.5A
+V5	5V	3.75A
+V5SUS	5V	0.5A
+V12	12V	0.25A
+V12S	12V	0.25A

SCREW HOLE

5E BLOCK DIAGRAM



FILE LIST

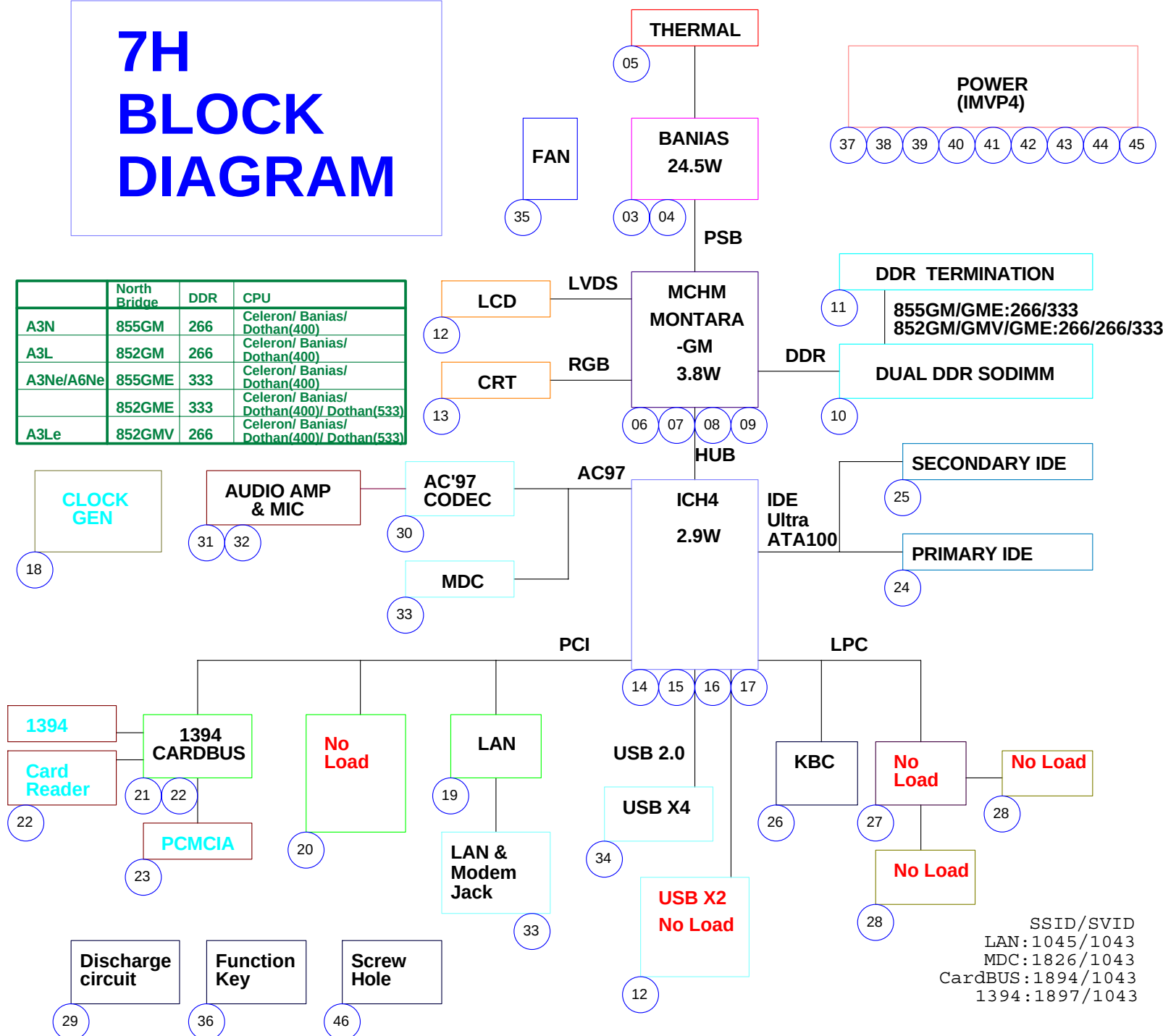
01

- 01_BLOCK DIAGRAM
- 02_POWER DIAGRAM
- 03_CPU-BANIAS(HOST)
- 04_CPU-BANIAS(PWR)
- 05_THERMAL
- 06_NB-MCHM(DDR)
- 07_NB-MCHM(HOST)
- 08_NB-MCHM(VGA)
- 09_NB-MCHM(PWR)
- 10_DUAL DDR SODIMM
- 11_DDR TERMINATION
- 12_LVDS & BACKLIGHT
- 13_CRT CONNECTOR
- 14_ICH4-M(HUB_PCI)
- 15_ICH4-M(H_U_IDE_PM)
- 16_ICH4-M(PWR)
- 17_ICH4-M_PULLUP
- 18_CLOCK-ICS950815
- 19_LAN-RTL8100CL
- 20_MINIPCI
- 21_CB1394-R5C593(1)
- 22_CB1394-R5C593(2)
- 23_PCPCIA SOCKET
- 24_IDE-HD
- 25_IDE-ODD
- 26_KBC-M38857
- 27_SIO-ITE7805
- 28_IR&LPT_PORT
- 29_DISCHARGE CIRCUIT
- 30_CODEC-ALC650
- 31_AUDIO AMP
- 32_MIC
- 33_MDC&RJ45&RJ11
- 34_USB
- 35_FAN&Audio DJ
- 36_FUNCTION KEY
- 37_PWR & RESET SEQ
- 38_VCORE
- 39_1.25V&1.8V
- 40_2.5V&1.5V&1.35V&1.05V
- 41_SYSTEM
- 42_LOAD SWITCH
- 43_CHARGER
- 44_PIC16C54
- 45_BATLOW/SD#
- 46_SCREW HOLE & M/B SETTING
- 47_REVISION(1)
- 48_REVISION(2)
- 49_REVISION(3)
- 50_REVISION(4)
- 51_BLOCK DIAGRAM 5E
- 52_BLOCK DIAGRAM 7H
- 53_Part Reference 1
- 54_Part Reference 2

SSID/SVID
LAN:1045/1043
MDC:1826/1043
CardBUS:1894/1043
1394:1897/1043

7H BLOCK DIAGRAM

	North Bridge	DDR	CPU
A3N	855GM	266	Celeron/ Banias/ Dothan(400)
A3L	852GM	266	Celeron/ Banias/ Dothan(400)
A3Ne/A6Ne	855GME	333	Celeron/ Banias/ Dothan(400)
	852GME	333	Celeron/ Banias/ Dothan(400)/ Dothan(533)
A3Le	852GMV	266	Celeron/ Banias/ Dothan(400)/ Dothan(533)



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BLOCK DIAGRAM